

# Symmetric exclusion versus independent random walks via the Octopus inequality

**Abstract.** We prove that the total-variation mixing time of the symmetric exclusion process is at most a universal constant times that of the same number of independent random walks, resolving a longstanding conjecture of Oliveira. The comparison holds on every finite connected weighted graph, uniformly in the number of particles, and extends to  $L^2$  mixing times. We also compare the Dirichlet forms of exclusion on the given graph and on the complete graph, with a coefficient that is optimal up to a universal constant.

## 1. Introduction

The symmetric exclusion process describes particles moving on a graph subject to the constraint that each vertex contains at most one particle: each particle attempts to follow the underlying random walk, but a jump to an occupied vertex is suppressed. A basic question is whether this interaction can increase the order of the mixing time. The occupation probabilities satisfy the single-particle heat equation, yet they alone do not determine convergence to equilibrium, which depends on the joint law of the occupation variables.

Oliveira<sup>1</sup> conjectured that the mixing time of exclusion is bounded by a universal constant times the natural independent-particle time scale. His general comparison incurred a logarithmic loss, and subsequent work proved the conjectured bounds for substantial classes of graphs and ranges of particle numbers. The question that remained was whether one constant suffices for all graphs, symmetric edge rates and particle numbers. In this paper, we prove that it does, both at fixed error and in Oliveira's formulation that retains the dependence on the error.

---

<sup>1</sup> R. I. Oliveira. 'Mixing of the symmetric exclusion processes in terms of the corresponding single-particle random walk'. In: *Ann. Probab.* 41.2 (2013), pp. 871–913, Sections 1.1 and 10.

The proof treats all particle densities within the same argument. At low density, particles may move for long periods before meeting, whereas at positive density many attempted jumps may be suppressed. We account for this interaction by estimating the harmonic components of the evolving density with respect to the stationary measure and proving a spectral lower bound that grows linearly with harmonic degree. The latter bound also gives a comparison with exclusion on the complete graph that is optimal up to a universal constant, from which we deduce logarithmic Sobolev inequalities and energy bounds for the spin-1/2 Heisenberg ferromagnet.

### 1.1 Oliveira's conjecture

Let  $V$  be a set of  $n \geq 2$  vertices, and let  $c_{xy} = c_{yx} \geq 0$  be edge rates, with  $c_{xx} = 0$ , such that the graph with edges  $\{x, y\}$  for which  $c_{xy} > 0$  is connected. We refer to this as connected positive support. For  $1 \leq k < n$ , we write  $EX(k)$  for exclusion on  $\binom{V}{k}$ ,  $RW(1)$  for the single-particle walk, and  $RW(k)$  for  $k$  independent labelled copies of that walk. The processes run in continuous time with the same edge rates and have uniform stationary measures on  $\binom{V}{k}$ ,  $V$  and  $V^k$ , respectively. For a process  $X$ , let  $t_{\text{mix}}^X(\varepsilon)$  be its worst-case total-variation mixing time at error  $\varepsilon$ , and write  $t_1(\varepsilon) = t_{\text{mix}}^{RW(1)}(\varepsilon)$ .

Early quantitative results exploited the geometry of particular graphs. Lee and Yau<sup>2</sup> obtained the order of the mixing time on a cycle when the numbers of particles and vacancies both grow at least as a fixed positive power of  $n$ , while Wilson<sup>3</sup> developed eigenfunction bounds for exclusion on a segment. Morris<sup>4</sup> introduced the chameleon process and determined the order of mixing on discrete tori of fixed dimension, including its dependence on the number of particles.

Oliveira<sup>5</sup> extended the use of the chameleon process to arbitrary connected weighted graphs, proving an upper bound of order  $t_1(1/4) \log(n/\varepsilon)$  for the exclusion mixing time, for  $0 < \varepsilon < 1/2$ . Thus the underlying graph enters the bound only through its number of vertices and the mixing time of one particle. His conjecture asks

<sup>2</sup> T.-Y. Lee and H.-T. Yau. 'Logarithmic Sobolev inequality for some models of random walks'. In: *Ann. Probab.* 26.4 (1998), pp. 1855–1873, Theorem 6.

<sup>3</sup> D. B. Wilson. 'Mixing times of lozenge tiling and card shuffling Markov chains'. In: *Ann. Appl. Probab.* 14.1 (2004). Author version cited, pp. 274–325. arXiv: math/0102193v2, Section 3.

<sup>4</sup> B. Morris. 'The mixing time for simple exclusion'. In: *Ann. Appl. Probab.* 16.2 (2006), pp. 615–635, Theorem 7.

<sup>5</sup> R. I. Oliveira. 'Mixing of the symmetric exclusion processes in terms of the corresponding single-particle random walk'. In: *Ann. Probab.* 41.2 (2013), pp. 871–913, Theorem 1.1.

whether the mixing time of the independent process already gives the correct upper bound, up to a universal constant.

There are two formulations of this question. At fixed error, his independent-particle heuristic<sup>6</sup> suggests that there should be a universal constant  $C$  such that

$$t_{\text{mix}}^{\text{EX}(k)}(1/4) \leq C t_{\text{mix}}^{\text{RW}(k)}(1/4). \quad (1.1)$$

To retain the dependence on the error, Oliveira<sup>7</sup> asks for

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq C t_1(\varepsilon/k) \quad (0 < \varepsilon < 1) \quad (1.2)$$

with a universal constant  $C$ . The time  $t_1(\varepsilon/k)$  arises naturally in the independent process: if each coordinate is within total variation distance  $\varepsilon/k$  of stationarity, then the product chain is within distance  $\varepsilon$ . Conversely, for  $0 < \varepsilon \leq 1/4$ , Proposition 2.4 shows that these two mixing times differ by at most a universal factor. Thus (1.2) implies the fixed-error comparison (1.1) while also specifying the dependence on the error. In both formulations, the constant is independent of the graph, the edge rates and the number of particles; in (1.2), it is also independent of the error. The error-dependent formulation replaces  $t_1(1/4) \log(n/\varepsilon)$  in Oliveira's general bound by  $t_1(\varepsilon/k)$ . This retains the dependence on the number of particles and can give a much smaller upper bound.

Hermon and Pymar<sup>8</sup> made substantial progress towards these comparisons. On regular graphs of degree  $d$ , with edge rates  $1/d$ , they obtained the general fixed-error bound  $O(\lambda^{-1} \log n \log \log n)$ , where  $\lambda$  is the single-particle spectral gap. Taking  $k \leq n/2$  by particle-hole symmetry, they proved the conjectured comparison for  $n^\beta \leq k \leq n^{1-\beta}$ , for each fixed  $\beta \in (0, 1/2)$ . For polynomially many particles, they also obtained the comparison when  $d \geq C \log_{n/k} n$ , or when  $\lambda^{-1} \geq c(\log n)^4$ , with this normalisation of the rates. Their spectral-profile estimate further refines the dependence on the number of particles throughout  $k \leq n^\delta$ , for each fixed  $\delta \in (0, 1)$ .

Comparisons for the labelled interchange process provide another approach to the question: the set of positions of any fixed

<sup>6</sup> R. I. Oliveira. 'Mixing of the symmetric exclusion processes in terms of the corresponding single-particle random walk'. In: *Ann. Probab.* 41.2 (2013), pp. 871–913, Section 1.1.

<sup>7</sup> R. I. Oliveira. 'Mixing of the symmetric exclusion processes in terms of the corresponding single-particle random walk'. In: *Ann. Probab.* 41.2 (2013), pp. 871–913, Section 10, third question.

<sup>8</sup> J. Hermon and R. Pymar. 'The exclusion process mixes (almost) faster than independent particles'. In: *Ann. Probab.* 48.6 (2020), pp. 3077–3123, Theorems 1.1–1.3 and the discussion following equation (5).

collection of labels evolves as exclusion. Hermon and Salez<sup>9</sup> proved the interchange mixing-time comparison on Cartesian products of connected graphs of a fixed size, including the hypercube. With unit edge rates, the interchange mixing time is of order  $\log|V|$ , and their comparison of Dirichlet forms gives exclusion estimates for every number of particles. Hermon and Pymar<sup>10</sup> obtained a direct comparison with independent walks for few labelled particles on vertex-transitive graphs of fixed degree  $d$  with equal edge rates. For fixed  $a \in (0, 1)$  and sufficiently large  $n$ , their comparison holds whenever  $3 \leq k \leq n^a$  and the error is at most  $1/4$ , with constants depending on  $a$  and  $d$ .

The relation between exclusion and independent walks is also central to the study of cutoff, as discussed in Salez's lecture notes.<sup>11</sup> For several classical geometries, a more precise description of convergence is available: Lacoïn and Leblond<sup>12</sup> proved cutoff on the complete graph, and Lacoïn proved cutoff on the segment<sup>13</sup> and determined the cutoff profile on the circle with half the vertices occupied.<sup>14</sup> More recently, Hermon, Kozma and Salez<sup>15</sup> proved exclusion cutoff under assumptions on the particle density and the single-particle return probabilities. Under the same return-probability assumption, their comparison of Dirichlet forms is asymptotically sharp at low density. Their heat-kernel comparison for interchange on arbitrary finite weighted graphs<sup>16</sup> is

- 
- <sup>9</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Sections 2.1 and 2.3.
- <sup>10</sup> J. Hermon and R. Pymar. *A direct comparison between the mixing time of the interchange process with "few" particles and independent random walks*. Preprint. 2021. arXiv: 2105.13486v1, Theorem 1.1 and the discussion following it.
- <sup>11</sup> J. Salez. *Modern aspects of Markov chains: entropy, curvature and the cutoff phenomenon*. Saint-Flour lecture notes. 2025. arXiv: 2508.21055v1, Section 2.3, pp. 27–28.
- <sup>12</sup> H. Lacoïn and R. Leblond. 'Cutoff phenomenon for the simple exclusion process on the complete graph'. In: *ALEA Lat. Am. J. Probab. Math. Stat.* 8 (2011), pp. 285–301, Theorem 1.1.
- <sup>13</sup> H. Lacoïn. 'Mixing time and cutoff for the adjacent transposition shuffle and the simple exclusion'. In: *Ann. Probab.* 44.2 (2016), pp. 1426–1487, Theorem 2.4.
- <sup>14</sup> H. Lacoïn. 'The cutoff profile for the simple exclusion process on the circle'. In: *Ann. Probab.* 44.5 (2016), pp. 3399–3430, Theorem 2.1.
- <sup>15</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, Theorem 1 and Corollary 4.
- <sup>16</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, Proposition 3.

also an ingredient of our proof, through its projection to exclusion.

The comparison results above cover a broad range of densities and geometries. The remaining question is whether the comparison holds uniformly in the number of particles on every finite connected graph with arbitrary symmetric edge rates. Even regular expanders illustrate the distinction: with unit total jump rate, the general bound of Hermon and Pymar is  $O(\log n \log \log n)$ , whereas the conjectured bound is  $O(\log n)$  for a positive density of particles. For small numbers of particles, the dependence on  $k$  is equally significant and is expressed by the time  $t_1(\varepsilon/k)$ . We prove both conjectured comparisons in this generality.

## 1.2 The universal comparison

Our main theorem proves (1.2) for every finite connected weighted graph and every error in  $(0, 1)$ . Particle-hole symmetry allows us to replace the number of particles by  $r = \min\{k, n - k\}$ .

**Theorem 1.1.** Let  $c_{xy} = c_{yx} \geq 0$  be edge rates with connected positive support on a finite set  $V$  of  $n \geq 2$  vertices. Let  $1 \leq k < n$  and  $r = \min\{k, n - k\}$ . Then

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq 34 t_{\text{mix}}^{\text{RW}(1)}(\varepsilon/r) \quad (0 < \varepsilon < 1). \quad (1.3)$$

For  $0 < \varepsilon \leq 1/2$ ,

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq 25 t_{\text{mix}}^{\text{RW}(1)}(\varepsilon/r). \quad (1.4)$$

Both processes use the edge rates  $c_{xy}$ .

Writing  $d_{\text{EX}(k)}(t)$  and  $d_1(t)$  for the corresponding worst-case distances, we also obtain

$$d_{\text{EX}(k)}(34t) \leq \min\{1, rd_1(t)\} \quad (t \geq 0); \quad (1.5)$$

see Corollary 7.1. This compares the distances from stationarity at all times. For independent particles, we obtain the fixed-error bound (1.1) with constant  $8 + 24 \log 6 < 52$ , and with  $k$  walks replaced by  $r$  walks. More generally, for every  $0 < \varepsilon \leq 1/4$ , the exclusion mixing time is at most 100 times that of  $r$  independent walks at the same error; see Corollary 7.2.

A stronger formulation, stated explicitly by Hermon and Pymar<sup>17</sup>, asks for comparison with independent particles at the same error throughout  $(0, 1)$ . For arbitrary symmetric edge rates, this extension fails. In Section 8, we construct a family of connected

---

<sup>17</sup> J. Hermon and R. Pymar. 'The exclusion process mixes (almost) faster than independent particles'. In: *Ann. Probab.* 48.6 (2020), pp. 3077–3123, equation (2).

weighted graphs on four vertices for which the ratio of the two-particle exclusion and independent-particle mixing times at error  $4/5$  tends to infinity. Theorem 1.1, by contrast, compares exclusion at error  $\varepsilon$  with a single particle at error  $\varepsilon/r$ , and holds throughout  $(0,1)$ .

The comparison with independent particles also holds for  $L^2$  and uniform mixing times; see Corollary 7.3.

### 1.3 Dirichlet-form comparison

The relation between single-particle motion and the convergence of an interacting system places Oliveira's conjecture in a broader setting, discussed by Hermon and Salez<sup>18</sup>. Aldous' theorem<sup>19</sup> identifies the spectral gaps of interchange, exclusion and the single-particle walk, and hence their slowest exponential rates of decay. For exclusion started from a fixed configuration, the resulting mixing bound contains  $\log \binom{n}{k}$ , the logarithm of the number of configurations. To obtain the independent-particle time scale, we use more of the spectrum: we prove a lower bound on each harmonic subspace and estimate the corresponding components of the evolving density. The spectral bound also yields the following comparison of Dirichlet forms.

Let  $A$  and  $H_k$  be the Laplacians of  $RW(1)$  and  $EX(k)$ , respectively, and let  $\lambda = \text{gap}(A)$ . Write  $H_k^{\text{comp}}$  for the exclusion Laplacian on the complete graph on  $V$  with every edge rate equal to one. These Laplacians are the negatives of the corresponding Markov generators. For self-adjoint operators,  $S \succeq T$  means that  $S - T$  is positive semidefinite. Both exclusion Laplacians act on functions on  $\binom{V}{k}$  with the uniform inner product.

**Theorem 1.2.** Let  $c_{xy} = c_{yx} \geq 0$  have connected positive support on  $V$ , and let  $n = |V| \geq 2$ . Let  $\lambda$  be the spectral gap of the single-particle walk with these rates. For  $1 \leq k < n$ , let  $H_k$  be the exclusion Laplacian with rates  $c_{xy}$ , and let  $H_k^{\text{comp}}$  be the exclusion Laplacian on  $V$  with unit rates on every edge. Then

$$H_k \succeq \frac{\lambda}{4n \log 6} H_k^{\text{comp}}. \quad (1.6)$$

<sup>18</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Section 1.2.

<sup>19</sup> P. Caputo, T. M. Liggett and T. Richthammer. 'Proof of Aldous' spectral gap conjecture'. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Theorem 1.1 and Section 4.1.1.

The coefficient has the best possible order. Indeed, on the mean-zero single-particle subspace, the complete-graph Laplacian is  $n\mathbf{I}$ , so no coefficient in such a comparison can exceed  $\lambda/n$ .

Diaconis and Saloff-Coste<sup>20</sup> compared exclusion with the Bernoulli-Laplace process by decomposing transpositions along paths. Alon and Kozma<sup>21</sup> subsequently used the octopus inequality to compare interchange with the complete graph. Hermon and Salez<sup>22</sup> conjectured that the single-particle relaxation time determines this comparison up to a universal constant. Theorem 1.2 proves the exclusion version of their conjecture for arbitrary symmetric edge rates.

To describe the spectral estimate, recall that functions on  $\binom{V}{k}$  decompose orthogonally into harmonic subspaces of degrees  $0 \leq j \leq r$ , each invariant under every transposition.<sup>23</sup> In Theorem 4.6, we show that the restriction of  $H_k$  to the subspace of degree  $j$  is bounded below by  $j\lambda/(4\log 6)$ . The linear dependence on  $j$ , together with bounds on the harmonic components of the density, gives the time scale  $t_1(\varepsilon/r)$ . Applied to the density of a point mass, the same spectral estimate gives the bound of order  $\lambda^{-1}\log(n/\varepsilon)$  in Corollary 7.4. This removes the  $\log\log n$  factor from the general regular-graph bound of Hermon and Pymar and, on regular expanders with unit total jump rate, gives an  $O(\log n)$  bound for every number of particles.

Hermon and Pymar<sup>24</sup> also relate Aldous' spectral-gap identity to a comparison of the distances from equilibrium of interchange and independent particles at the same time and from the same initial configuration. They also discuss an  $L^2$  version of Oliveira's comparison, which we establish in Corollary 7.3. In our argument, the additional spectral information has a representation-theoretic interpretation. The harmonic spaces afford the irreducible

<sup>20</sup> P. Diaconis and L. Saloff-Coste. 'Comparison theorems for reversible Markov chains'. In: *Ann. Appl. Probab.* 3.3 (1993), pp. 696-730, Section 3, Lemmas 3.1-3.2.

<sup>21</sup> G. Alon and G. Kozma. 'Comparing with octopi'. In: *Ann. Inst. Henri Poincaré Probab. Stat.* 56.4 (2020). Author version cited, pp. 2672-2685. arXiv: 1811.10537v1, Theorem 1.

<sup>22</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84-98. arXiv: 1905.02146v2, Section 2.1, Conjecture 2.

<sup>23</sup> Y. Filmus. 'An orthogonal basis for functions over a slice of the Boolean hypercube'. In: *Electron. J. Combin.* 23.1, 1.23 (2016). 27 pp., Theorem 15.

<sup>24</sup> J. Hermon and R. Pymar. 'The exclusion process mixes (almost) faster than independent particles'. In: *Ann. Probab.* 48.6 (2020), pp. 3077-3123, Section 1.5.

representations indexed by  $(\mathbf{n}-\mathbf{j}, \mathbf{j})$ <sup>25</sup>; for  $\mathbf{j} \geq \mathbf{1}$ , Aldous' theorem bounds the least eigenvalue of  $H_{\mathbf{k}}$  on each of these spaces below by  $\lambda$ , whereas our bound is proportional to  $\mathbf{j}\lambda$ . This gives a lower bound that depends on the second row of the indexing partition. Comparisons of least eigenvalues across irreducible representations are also the subject of the Aldous order introduced by Alon and Kozma<sup>26</sup>.

#### 1.4 Further consequences

Theorem 1.2 allows us to apply functional inequalities for the Bernoulli-Laplace process to exclusion on an arbitrary weighted graph. Combining it with the logarithmic Sobolev inequality of Lee and Yau<sup>27</sup>, we obtain a lower bound of order  $\lambda/\log(\mathbf{n}^2/[\mathbf{k}(\mathbf{n}-\mathbf{k})])$  for the exclusion logarithmic Sobolev constant; see Corollary 7.5. Hence, when both particles and vacancies occupy a positive proportion of the vertices, this constant is comparable to  $\lambda$ . Salez<sup>28</sup> derived the corresponding conclusion for balanced colour classes under the conjectured interchange comparison. Our theorem proves it unconditionally for two colours and arbitrary symmetric edge rates.

The correspondence between symmetric exclusion and the spin-1/2 Heisenberg ferromagnet<sup>29</sup> gives another consequence of the spectral bound. Harmonic degree is the deficit of the total spin from its maximum, so we obtain an energy lower bound proportional to this deficit times the single-particle gap. The same operator occurs in quantum-cut sparsification, where Basu, Brakensiek, Kothari and Putterman<sup>30</sup> use a complete-graph comparison to control energies on expanders. Our theorem proves that operator inequality on every

---

<sup>25</sup> Y. Filmus. 'An orthogonal basis for functions over a slice of the Boolean hypercube'. In: *Electron. J. Combin.* 23.1, 1.23 (2016). 27 pp., Section 4, following Lemma 18.

<sup>26</sup> G. Alon and G. Kozma. 'Ordering the representations of  $S_n$  using the interchange process'. In: *Canad. Math. Bull.* 56.1 (2013), pp. 13-30, Section 1.

<sup>27</sup> T.-Y. Lee and H.-T. Yau. 'Logarithmic Sobolev inequality for some models of random walks'. In: *Ann. Probab.* 26.4 (1998), pp. 1855-1873, Theorem 5.

<sup>28</sup> J. Salez. 'A sharp log-Sobolev inequality for the multislice'. In: *Ann. H. Lebesgue* 4 (2021), pp. 1143-1161, Corollary 2.6 and the discussion following it.

<sup>29</sup> B. Nachtergaele and S. Starr. 'Ordering of energy levels in Heisenberg models and applications'. In: *Mathematical Physics of Quantum Mechanics*. Ed. by J. Asch and A. Joye. Vol. 690. Lecture Notes in Phys. Author version cited. Berlin: Springer, 2006, pp. 149-170. arXiv: math-ph/0503056v1, Section 5.3, Proposition 2.

<sup>30</sup> A. Basu et al. *Quantum cut sparsifiers*. Preprint. 2026. arXiv: 2606.09728v1, Theorem 5.5.

finite connected weighted graph, with a coefficient that is optimal up to a universal constant. We give the precise statements and normalisations in Section 7.6.

### 1.5 Proof and organisation

By particle-hole symmetry, it suffices to consider  $k \leq n/2$ . The proof has two main parts: a spectral estimate on the harmonic subspaces and a bound on the components of the evolving density in those subspaces. The first begins with the octopus inequality of Caputo, Liggett and Richthammer<sup>31</sup>, which arose from the network-reduction approach to Aldous' conjecture, developed independently by Dieker<sup>32</sup> and by Caputo, Liggett and Richthammer. Alon and Kozma<sup>33</sup> compared interchange Laplacians by summing the octopus inequality over the vertices and iterating the resulting comparison. Hermon and Salez<sup>34</sup> used this argument in convolution form. We use the heat-kernel comparison of Hermon, Kozma and Salez<sup>35</sup>, which follows from the same octopus argument. Another application of the octopus inequality is Chen's moving particle lemma<sup>36</sup>, which bounds the mean squared change of a function under a transposition by the effective resistance between its endpoints times the exclusion Dirichlet form.

For the linear map  $\Phi_k$  from single-particle matrices to exclusion operators defined in Section 3, the heat-kernel comparison reads

$$\Phi_k(I - e^{-tA}) \preceq tH_k \quad (t \geq 0). \quad (1.7)$$

Thus the exclusion Laplacian with heat-kernel edge rates is bounded above, in the order of quadratic forms, by  $tH_k$ .

To use this comparison, we extend a degree- $j$  harmonic function by zero from tuples with pairwise distinct coordinates to  $V^j$ . The

- 
- <sup>31</sup> P. Caputo, T. M. Liggett and T. Richthammer. 'Proof of Aldous' spectral gap conjecture'. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Theorem 2.3.
- <sup>32</sup> A. B. Dieker. 'Interlacings for random walks on weighted graphs and the interchange process'. In: *SIAM J. Discrete Math.* 24.1 (2010). Author version cited, pp. 191–206. arXiv: 0906.1716v3, Conjecture 1 and Section 4.
- <sup>33</sup> G. Alon and G. Kozma. 'Comparing with octopi'. In: *Ann. Inst. Henri Poincaré Probab. Stat.* 56.4 (2020). Author version cited, pp. 2672–2685. arXiv: 1811.10537v1, proof of Theorem 2, equations (5)–(8).
- <sup>34</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Section 3.3, Lemma 3 and equation (60).
- <sup>35</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, Proposition 3.
- <sup>36</sup> J. P. Chen. 'The moving particle lemma for the exclusion process on a weighted graph'. In: *Electron. Commun. Probab.* 22, 47 (2017). Author version cited. arXiv: 1606.01577v4, Theorem 1.

extension has mean zero on each coordinate fibre, so applying the single-particle Poincaré inequality on every fibre and summing gives a factor  $j$ . The difference between the product-chain and exclusion Rayleigh quotients is an explicit collision term. We bound it using the constant and mean-zero parts of the heat kernel at time  $(\log 6)/\lambda$ ; together with (1.7), this gives the required spectral lower bound.

For the second part, we first allow the process to evolve for time  $t$  and then condition on configurations avoiding vertices whose occupation probabilities exceed  $2k/n$ . The discarded probability is at most  $2kd_1(t)$ . Applying the classical negative correlation inequality to the unconditioned law, and then dividing by the probability of the conditioning event, gives bounds on the inclusion probabilities of the conditional law. Using an averaging operator, we bound the squared  $L^2$  norm of the degree- $j$  harmonic component of the conditional density by  $p^{-1}(4k)^j/j!$ , where  $p$  is the probability of the conditioning event. We can now apply the spectral bound on each harmonic subspace and sum over the degrees to obtain the mixing estimate. A separate two-particle argument completes the proof for errors close to one.

In Section 2, we fix the Markov-chain conventions and compare single-particle and product mixing times. In Section 3, we present the heat-kernel comparison, and in Section 4, we establish the spectral bound on each harmonic subspace. In Section 5, we estimate the harmonic components of the density and derive a two-time mixing bound. In Section 6, we prove Theorem 1.1. In Section 7, we prove Theorem 1.2 and derive comparisons with independent particles in total variation,  $L^2$  and uniform distance, together with logarithmic Sobolev inequalities and spin-system consequences. In Section 8, we give the four-vertex counterexample.

**Research process and use of generative AI.** The starting point for this study was the analogy drawn by Hermon and Salez<sup>37</sup> between Oliveira's conjecture and Aldous' spectral gap conjecture, which suggested to me the use of the octopus inequality. During this study, I used ChatGPT (versions 5.4, 5.5 and 5.6, accessed through ChatGPT Plus) to help me understand background material and identify standard terminology. ChatGPT also suggested an early argument for the heat-kernel comparison. This argument is superseded by the more general interchange comparison of Hermon,

---

<sup>37</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Section 1.2.

Kozma and Salez, which supplies the heat-kernel result used in this paper. I am grateful to them for this result. I also used ChatGPT to assist with checking references and for proofreading and improving the English. I take full responsibility for the mathematical content and the writing of the manuscript.

## 2. Chains and the single-particle time scale

In this section, we fix our notation and compare the mixing time of independent particles with that of a single walk at a smaller error. The latter comparison will allow us to express the main estimates in terms of the single-particle walk.

Throughout,  $V$  is a set of  $n \geq 2$  vertices, the rates satisfy  $c_{xy} = c_{yx} \geq 0$  and  $c_{xx} = 0$ , and the graph formed by the edges of positive rate is connected. A fixed ordering of  $V$  allows us to sum over unordered pairs by writing  $x < y$ . All logarithms are natural.

### 2.1 Laplacians, stationary measures and mixing times

The Laplacian of the single-particle walk is

$$(Af)(x) = \sum_{y \in V} c_{xy} (f(x) - f(y)). \quad (2.1)$$

The corresponding walk has transition kernel  $P_t = e^{-tA}$  and stationary distribution  $\pi = \text{Unif}(V)$ . Denote the smallest positive eigenvalue of  $A$  by  $\lambda$ , and write  $t_{\text{rel}} = \lambda^{-1}$  for its relaxation time. For  $0 \leq r \leq n$ , let  $\Omega_r = \binom{V}{r}$  and  $\pi_r = \text{Unif}(\Omega_r)$ . We use  $\tau_{xy}$  for both the transposition of  $x$  and  $y$  and its action on functions,  $(\tau_{xy}f)(B) = f(\tau_{xy}B)$ . The exclusion Laplacian is

$$H_r = \sum_{x < y} c_{xy} (I - \tau_{xy}) \quad \text{on } L^2(\pi_r), \quad (2.2)$$

with transition kernel  $Q_t^{(r)} = e^{-tH_r}$ . Thus  $A$  and  $H_r$  are the negatives of the corresponding Markov generators.<sup>38</sup> A transposition of two occupied or two empty vertices leaves the configuration unchanged; thus each particle jumps from  $x$  to an unoccupied vertex  $y$  at rate  $c_{xy}$ , in agreement with the time normalisation of the single walk.

For a probability measure  $\sigma$  on a finite set, write  $\langle f, g \rangle_\sigma = \sum_x \sigma(x) f(x) g(x)$  and  $\|f\|_{2,\sigma}^2 = \langle f, f \rangle_\sigma$ . For probability measures  $\mu, \nu$

<sup>38</sup> P. Caputo, T. M. Liggett and T. Richthammer. ‘Proof of Aldous’ spectral gap conjecture’. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Section 1.1.

on the same set, let  $\|\mu - \nu\|_{TV} = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$ ; we use the same formula for signed measures of total mass zero. The worst-case distances from stationarity are

$$d_1(t) = \max_{x \in V} \|P_t(x, \cdot) - \pi\|_{TV}, \quad (2.3)$$

$$d_{EX(r)}(t) = \max_{B \in \Omega_r} \|Q_t^{(r)}(B, \cdot) - \pi_r\|_{TV}, \quad (2.4)$$

$$d_{RW(r)}(t) = \max_{x \in V^r} \|P_t^{\otimes r}(x, \cdot) - \pi^{\otimes r}\|_{TV}. \quad (2.5)$$

The maximum in (2.5) includes tuples with repeated vertices. For each chain  $X$ , we define its mixing time  $t_{mix}^X(\varepsilon)$  to be the infimum of the times at which its worst-case distance is at most  $\varepsilon$ , and abbreviate  $t_1(\varepsilon) = t_{mix}^{RW(1)}(\varepsilon)$ .<sup>39</sup>

Since the complement map commutes with every transposition and sends  $\pi_r$  to  $\pi_{n-r}$ , particle-hole symmetry gives

$$d_{EX(r)}(t) = d_{EX(n-r)}(t) \quad (1 \leq r < n, \ t \geq 0). \quad (2.6)$$

Aldous' spectral-gap conjecture, posed in the early 1990s, asserted that interchange and the single-particle walk have the same gap. Handjani and Jungreis<sup>40</sup> proved the identity for weighted trees, while Koma and Nachtergaele<sup>41</sup> obtained related identities for spin chains. The conjecture was proved on every finite connected weighted graph by Caputo, Liggett and Richthammer<sup>42</sup>; we shall use the following consequence for exclusion<sup>43</sup>.

**Theorem 2.1** (Exclusion spectral gap). Let  $1 \leq r < n$ . Then the smallest positive eigenvalue of  $H_r$  is  $\lambda$ . In particular, for every real function  $f$  on  $\Omega_r$ ,

$$\langle f, H_r f \rangle_{\pi_r} \geq \lambda \text{Var}_{\pi_r}(f). \quad (2.7)$$

<sup>39</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Proposition 4.2 and Section 20.2, equation (20.9).

<sup>40</sup> S. Handjani and D. Jungreis. 'Rate of convergence for shuffling cards by transpositions'. In: *J. Theoret. Probab.* 9.4 (1996), pp. 983–993.

<sup>41</sup> T. Koma and B. Nachtergaele. 'The spectral gap of the ferromagnetic XXZ chain'. In: *Lett. Math. Phys.* 40.1 (1997). Author version cited, pp. 1–16. arXiv: cond-mat/9512120v1, Proposition 2.

<sup>42</sup> P. Caputo, T. M. Liggett and T. Richthammer. 'Proof of Aldous' spectral gap conjecture'. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Theorem 1.1.

<sup>43</sup> P. Caputo, T. M. Liggett and T. Richthammer. 'Proof of Aldous' spectral gap conjecture'. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Section 4.1.1, equation (4.1).

## 2.2 Separation and contraction

For a Markov semigroup  $K_t$  on a finite state space with strictly positive stationary distribution  $\sigma$ , its separation distance and total-variation contraction coefficient are

$$s_K(t) = 1 - \min_{x,y} \frac{K_t(x,y)}{\sigma(y)}, \quad \bar{d}_K(t) = \max_{x,y} \|K_t(x, \cdot) - K_t(y, \cdot)\|_{TV}. \quad (2.8)$$

For the discrete-time definitions, see Levin and Peres.<sup>44</sup> Write  $d_K(t)$  for its worst-case distance from  $\sigma$ . For the single-particle walk, we write  $s_1(t)$  and  $\bar{d}_1(t)$  for the corresponding separation distance and contraction coefficient. Since  $K_t(x, \cdot) \geq (1 - s_K(t))\sigma$  for every  $x$ , we have  $d_K(t) \leq s_K(t)$ . The following estimates describe how these distances behave under composition and products.

**Lemma 2.2.** Let  $K_t$  be a Markov semigroup on a finite state space with strictly positive stationary distribution. Then, for  $t, u \geq 0$ ,

$$\bar{d}_K(t) \leq 2d_K(t), \quad d_K(t+u) \leq d_K(t)\bar{d}_K(u). \quad (2.9)$$

If  $K_t$  is reversible, then

$$s_K(2t) \leq 2d_K(t). \quad (2.10)$$

For every integer  $r \geq 1$ , the product of  $r$  independent copies of the single-particle walk satisfies

$$1 - s_{RW(r)}(t) = (1 - s_1(t))^r. \quad (2.11)$$

*Proof.* The contraction estimates are standard properties of total variation.<sup>45</sup>

For a reversible semigroup, put  $f(z) = K_t(x, z)/\sigma(z)$  and  $g(z) = K_t(y, z)/\sigma(z)$ . Then

$$\begin{aligned} \frac{K_{2t}(x, y)}{\sigma(y)} &= \sum_z \sigma(z) f(z) g(z) \\ &\geq \sum_z \sigma(z) \min\{f(z), 1\} \min\{g(z), 1\} \\ &\geq 1 - \|K_t(x, \cdot) - \sigma\|_{TV} - \|K_t(y, \cdot) - \sigma\|_{TV}, \end{aligned}$$

<sup>44</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, equations (6.6)–(6.7) and (4.23).

<sup>45</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Lemma 4.10 and Exercise 4.3.

where the last inequality follows from  $ab \geq a + b - 1$  for  $a, b \in [0, 1]$ . Taking the minimum over  $x, y$  gives (2.10). Finally, the transition density of the product chain with respect to its stationary measure factors over the coordinates. Its minimum is therefore the product of the single-coordinate minima. This proves (2.11).  $\square$

We first record the decay of the single-particle distance at successive multiples of its mixing time.

**Lemma 2.3.** Let  $0 < \delta < 1/2$  and  $T = t_1(\delta)$ . Then, for every integer  $m \geq 1$ ,

$$d_1(mT) \leq \delta(2\delta)^{m-1}, \quad \lambda T \geq \log \frac{1}{2\delta}. \quad (2.12)$$

*Proof.* Continuity gives  $d_1(T) \leq \delta$ , so iteration of (2.9) proves the first bound. For the second, apply the standard spectral lower bound at  $t = T^{46}$

$$d_1(t) \geq \frac{1}{2}e^{-\lambda t}. \quad (2.13)$$

$\square$

We now compare the mixing time of independent walks with that of a single walk. The usual product bound gives one direction; separation will give the converse comparison up to a universal constant.

**Proposition 2.4.** Let  $k \geq 1$  be an integer and  $0 < \varepsilon \leq 1/4$ . Then

$$t_{\text{mix}}^{\text{RW}(k)}(\varepsilon) \leq t_1(\varepsilon/k) \leq 4t_{\text{mix}}^{\text{RW}(k)}(\varepsilon). \quad (2.14)$$

*Proof.* The first inequality follows from the product bound<sup>47</sup>,  $d_{\text{RW}(k)}(t) \leq kd_1(t)$ . To prove the reverse bound, let  $T = t_{\text{mix}}^{\text{RW}(k)}(\varepsilon)$ . The separation estimates in Lemma 2.2 give

$$d_1(2T) \leq 1 - (1 - 2\varepsilon)^{1/k} \leq \frac{-\log(1 - 2\varepsilon)}{k} \leq \frac{4(\log 2)\varepsilon}{k}.$$

Here the last inequality follows from convexity of  $-\log(1 - 2x)$  on  $[0, 1/4]$ . On the other hand, projection onto one coordinate gives  $d_1(T) \leq \varepsilon$  and hence  $\bar{d}_1(T) \leq 2\varepsilon$ . Two further applications of (2.9) now give

$$d_1(4T) \leq \frac{4(\log 2)\varepsilon}{k}(2\varepsilon)^2 \leq \frac{\varepsilon}{k}. \quad \square$$

<sup>46</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Lemma 20.11, equation (20.34).

<sup>47</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Exercise 4.4.

For use in the fixed-error comparison, we record the following consequences at error  $1/4$ .

**Proposition 2.5.** Let  $1 \leq k \leq n$  and  $T = t_{\text{mix}}^{\text{RW}(k)}(1/4)$ . Then

$$T \leq t_1(1/(4k)) \leq 4T, \quad (2.15)$$

$$d_1(8T) \leq \frac{1}{64k}, \quad T \geq \frac{\log(2k)}{4\lambda}. \quad (2.16)$$

*Proof.* Proposition 2.4 gives (2.15) and  $d_1(4T) \leq 1/(4k)$ . Since  $\bar{d}_1(T) \leq 2d_1(T) \leq 1/2$ , four applications of (2.9) then give  $d_1(8T) \leq 1/(64k)$ . Finally, comparing (2.13) at time  $4T$  with  $d_1(4T) \leq 1/(4k)$  yields the lower bound on  $T$ .  $\square$

### 3. A heat-kernel comparison

The heat-kernel comparison below is the exclusion case of the interchange comparison proved by Hermon, Kozma and Salez.<sup>48</sup> We include a proof in our matrix notation, based on summing the octopus inequality and iterating the resulting comparison, as in Alon and Kozma<sup>49</sup> and Hermon and Salez.<sup>50</sup>

#### 3.1 Exclusion operators and the octopus inequality

For self-adjoint operators on the same real inner product space, write  $S \preceq T$  if  $\langle f, (T-S)f \rangle \geq 0$  for every  $f$ . We denote the all-ones vector by  $\mathbf{1}$ .

**Definition 3.1.** Let  $M$  be a real symmetric matrix indexed by  $V$  with  $M\mathbf{1} = 0$ , and let  $0 \leq r \leq n$ . Define the operator

$$\Phi_r(M) = - \sum_{x < y} M_{xy} (I - \tau_{xy}) \quad \text{on } L^2(\pi_r). \quad (3.1)$$

The map  $\Phi_r$  is linear and sends a graph Laplacian  $M$  to the exclusion Laplacian with rates  $-M_{xy}$ . In particular,  $\Phi_r(A) = H_r$ . The comparison uses the fact that  $\Phi_r$  sends the square of a graph Laplacian to a positive semidefinite operator; we derive this from the octopus inequality below.

<sup>48</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, Proposition 3.

<sup>49</sup> G. Alon and G. Kozma. 'Comparing with octopi'. In: *Ann. Inst. Henri Poincaré Probab. Stat.* 56.4 (2020). Author version cited, pp. 2672–2685. arXiv: 1811.10537v1, proof of Theorem 2, equations (5)–(8).

<sup>50</sup> J. Hermon and J. Salez. 'The interchange process on high-dimensional products'. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Section 3.3, Lemma 3 and equation (60).

The octopus inequality of Caputo, Liggett and Richthammer<sup>51</sup> applies to functions of a permutation. Restricting to functions that depend only on the image of a fixed  $r$ -element set gives the following form on subsets; zero weights are allowed by continuity. For a vertex  $z$  and non-negative weights  $b_{zx}$ , write  $d_z = \sum_{x \neq z} b_{zx}$ .

**Theorem 3.2** (Octopus inequality). Let  $z \in V$ , let  $b_{zx} \geq 0$  for  $x \neq z$ , and let  $0 \leq r \leq n$ . Then, on  $L^2(\pi_r)$ ,

$$d_z \sum_{x \neq z} b_{zx} (I - \tau_{zx}) \succeq \sum_{\substack{x < y \\ x, y \neq z}} b_{zx} b_{zy} (I - \tau_{xy}). \quad (3.2)$$

**Lemma 3.3.** Let  $K$  be the Laplacian of a symmetric weighted graph on  $V$ , not necessarily connected. Then

$$\Phi_r(K^2) \succeq 0 \quad (0 \leq r \leq n). \quad (3.3)$$

*Proof.* Write  $K_{xy} = -b_{xy}$  for  $x \neq y$  and  $K_{xx} = d_x = \sum_{y \neq x} b_{xy}$ . Summing (3.2) over all centres  $z \in V$  gives  $\Phi_r(K^2) \succeq 0$ . Indeed, the row sums of  $K^2$  vanish, and the coefficient of  $I - \tau_{xy}$  in the sum is

$$(d_x + d_y)b_{xy} - \sum_{z \notin \{x, y\}} b_{xz} b_{zy} = -(K^2)_{xy}. \quad (3.4) \quad \square$$

### 3.2 Comparison by dyadic subdivision

We can now use the semigroup identity to compare successive time scales, and then pass to the Laplacian as the time step tends to zero. The following is the comparison of Hermon, Kozma and Salez in our normalisation. Indeed, choose  $R \geq \max_x A_{xx} > 0$  and put  $P = I - A/R$ , a symmetric stochastic matrix. The continuous-time walk with generator  $P - I$  has heat kernel  $e^{-tA}$  at time  $Rt$ , and the corresponding exclusion Laplacian is  $H_r/R$ . Applying their comparison at time  $Rt$  gives the stated inequality. The change from counting measure to  $\pi_r$  multiplies both forms by the same constant.

**Theorem 3.4.** Let  $A$  be the single-particle Laplacian in (2.1). Then, for every  $t \geq 0$  and  $0 \leq r \leq n$ ,

$$\Phi_r(I - e^{-tA}) \preceq tH_r. \quad (3.5)$$

*Proof.* Let  $K_t = I - P_t$ . Since  $P_t$  is a symmetric stochastic matrix,  $K_t$  is a graph Laplacian with edge rates  $P_t(x, y)$  for  $x \neq y$ . The semigroup identity gives

$$K_{2t} = 2K_t - K_t^2. \quad (3.6)$$

<sup>51</sup> P. Caputo, T. M. Liggett and T. Richthammer. ‘Proof of Aldous’ spectral gap conjecture’. In: *J. Amer. Math. Soc.* 23.3 (2010). Author version cited, pp. 831–851. arXiv: 0906.1238v4, Theorem 2.3.

By linearity of  $\Phi_r$  and Lemma 3.3, we obtain  $\Phi_r(K_{2t}) \preceq 2\Phi_r(K_t)$  and hence, by iteration,

$$\Phi_r(I - P_t) \preceq 2^m \Phi_r(I - P_{t/2^m}) \quad (m \geq 0). \quad (3.7)$$

Now  $2^m(I - e^{-tA/2^m}) \rightarrow tA$  in matrix norm. Since  $\Phi_r$  is continuous and the positive semidefinite cone is closed, passage to the limit proves (3.5).  $\square$

We shall apply Theorem 3.4 at a fixed multiple of  $t_{\text{rel}}$ , using an operator-norm bound on the heat kernel.

#### 4. Harmonic decomposition and spectral bounds

We decompose functions on  $\Omega_k$  into orthogonal harmonic subspaces and prove a spectral lower bound proportional to the degree on each subspace. The decomposition is independent of the edge rates; the spectral estimate follows by applying the heat-kernel comparison to the zero extension of a harmonic function. Write  $(a)_j = a(a-1)\cdots(a-j+1)$ , with  $(a)_0 = 1$ .

##### 4.1 Raising and lowering operators

For  $0 \leq r < n$ , the raising and lowering operators between the layers  $\Omega_r$  and  $\Omega_{r+1}$  are

$$(R_r f)(B) = \sum_{\substack{S \subset B \\ |S|=r}} f(S), \quad (D_{r+1} g)(S) = \sum_{x \notin S} g(S \cup \{x\}).$$

Here  $f$  is a function on  $\Omega_r$  and  $g$  is a function on  $\Omega_{r+1}$ . We set  $D_0 = 0$  and interpret  $R_{-1}D_0$  as zero. Throughout this subsection, adjoints and norms are taken with respect to counting measure unless stated otherwise; in particular,  $R_r^* = D_{r+1}$ .

**Definition 4.1.** For  $0 \leq j \leq n/2$ , the space of harmonic functions on  $\Omega_j$  is  $\mathcal{H}_j = \ker D_j$ . For  $j \leq k \leq n$ , the inclusion operator  $U_{j,k}$  is given by

$$(U_{j,k} h)(B) = \sum_{\substack{S \subset B \\ |S|=j}} h(S), \quad (4.1)$$

and  $\mathcal{H}_j^{(k)} = U_{j,k} \mathcal{H}_j$ . We call  $\mathcal{H}_j^{(k)}$  the harmonic subspace of degree  $j$  on  $\Omega_k$ .

In particular,  $\mathcal{H}_0 = \mathbb{R}^{\Omega_0}$ . For  $j \geq 1$ , a function  $h$  belongs to  $\mathcal{H}_j$  precisely when

$$\sum_{x \notin S} h(S \cup \{x\}) = 0 \quad (S \in \Omega_{j-1}). \quad (4.2)$$

The inclusion operators intertwine the permutation actions on the respective layers, and repeated raising gives  $U_{j,k} = R_{k-1} \cdots R_j / (k-j)!$  for  $k > j$ . Under the identification of  $h$  with the multilinear polynomial  $\sum_{S \in \Omega_j} h(S) \prod_{x \in S} z_x$ , the condition  $D_j h = 0$  is the usual harmonicity condition for multilinear polynomials.<sup>52</sup>

The following lemma gives the standard decomposition<sup>53</sup> and records the adjoint and norm identities for our normalisation of  $U_{j,k}$ . The raising and lowering relations and the corresponding norm formula are treated by Srinivasan.<sup>54</sup> Hermon, Kozma and Salez<sup>55</sup> use the same decomposition in their spectral analysis of exclusion.

**Lemma 4.2.** Let  $0 \leq r < n$ . Then

$$D_{r+1} R_r - R_{r-1} D_r = (n - 2r)I. \quad (4.3)$$

For  $0 \leq j \leq k \leq n/2$  and  $h \in \mathcal{H}_j$ ,

$$U_{j,k}^* U_{j,k} h = \binom{n-2j}{k-j} h, \quad (4.4)$$

and consequently

$$\sum_{B \in \Omega_k} (U_{j,k} h)(B)^2 = \binom{n-2j}{k-j} \sum_{S \in \Omega_j} h(S)^2. \quad (4.5)$$

Moreover, for  $0 \leq k \leq n/2$ ,

$$L^2(\pi_k) = \bigoplus_{j=0}^k \mathcal{H}_j^{(k)}, \quad \dim \mathcal{H}_j^{(k)} = \binom{n}{j} - \binom{n}{j-1}, \quad (4.6)$$

where the sum is orthogonal and  $\binom{n}{-1} = 0$ .

*Proof.* To prove (4.3), compare the two compositions: the terms that replace one element of a set by another cancel, while the coefficients of the unchanged set are  $n-r$  and  $r$ , whose difference is  $n-2r$ .

<sup>52</sup> Y. Filmus. ‘An orthogonal basis for functions over a slice of the Boolean hypercube’. In: *Electron. J. Combin.* 23.1, 1.23 (2016). 27 pp., Definition 1.

<sup>53</sup> Y. Filmus. ‘An orthogonal basis for functions over a slice of the Boolean hypercube’. In: *Electron. J. Combin.* 23.1, 1.23 (2016). 27 pp., Lemma 2 and Theorem 15.

<sup>54</sup> M. K. Srinivasan. ‘Symmetric chains, Gelfand-Tsetlin chains, and the Terwilliger algebra of the binary Hamming scheme’. In: *J. Algebraic Combin.* 34.2 (2011). Author version cited, pp. 301–322. arXiv: 1001.0280v2, Section 1, Theorem 1.2 and equation (4).

<sup>55</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, Theorem 2.

For  $h \in \ker D_j$ , induction using (4.3) gives

$$D_{j+m} R_{j+m-1} \cdots R_j h = m(n-2j-m+1) R_{j+m-2} \cdots R_j h \quad (4.7)$$

for  $1 \leq m \leq n-j$ , with an empty product interpreted as the identity. Applying all  $m$  lowering operators to  $R_{j+m-1} \cdots R_j h$  therefore gives  $m!(n-2j)_m h$ . Taking account of the normalisation  $U_{j,j+m} = R_{j+m-1} \cdots R_j / m!$ , we obtain (4.4) and (4.5).

We next determine the dimensions of the harmonic subspaces. For  $r < n/2$ , the commutator gives  $\|R_r f\|^2 = \|D_r f\|^2 + (n-2r)\|f\|^2$ , so  $R_r$  is injective and  $D_{r+1}$  is surjective. It follows that  $\dim \mathcal{H}_j = \binom{n}{j} - \binom{n}{j-1}$ , while (4.4) shows that  $U_{j,k}$  is injective on  $\mathcal{H}_j$  in the stated range.

It remains to prove orthogonality. For  $i < j$ , let  $h \in \mathcal{H}_i$  and  $g \in \mathcal{H}_j$ . Moving the raising operators in  $U_{j,k}$  across the inner product and applying (4.7) reduces  $\langle U_{i,k} h, U_{j,k} g \rangle$  to a scalar multiple of  $\langle U_{i,j} h, g \rangle$ , which vanishes because  $U_{i,j} h \in \text{im } R_{j-1}$  and  $g \in \ker D_j$ . The dimensions sum to  $\binom{n}{k}$ , and counting measure and  $\pi_k$  give the same orthogonality. Hence the harmonic subspaces form the claimed orthogonal decomposition.  $\square$

The next lemma allows us to estimate the exclusion Laplacian on each harmonic subspace by working on the layer where that subspace first appears.

**Lemma 4.3.** Let  $0 \leq j \leq k \leq n/2$ . Then  $\mathcal{H}_j$  and  $\mathcal{H}_j^{(k)}$  are invariant under  $H_j$  and  $H_k$ , respectively, and

$$H_k U_{j,k} = U_{j,k} H_j. \quad (4.8)$$

For every non-zero  $h \in \mathcal{H}_j$ ,

$$\frac{\langle U_{j,k} h, H_k U_{j,k} h \rangle_{\pi_k}}{\|U_{j,k} h\|_{2, \pi_k}^2} = \frac{\langle h, H_j h \rangle_{\pi_j}}{\|h\|_{2, \pi_j}^2}. \quad (4.9)$$

*Proof.* Both lowering and inclusion intertwine the permutation actions on their respective layers. They therefore intertwine the exclusion Laplacians, which proves invariance and (4.8). Moreover, by (4.4), inclusion multiplies the energy and squared norm in counting measure by the same factor  $\binom{n-2j}{k-j}$ . Their quotient is thus unchanged, and passing to uniform probability measures leaves it unchanged as well. This gives (4.9).  $\square$

Let  $H_k^{\text{comp}} = \sum_{x < y} (I - \tau_{xy})$  be the exclusion Laplacian on the complete graph with every edge rate equal to one. For this operator, the

harmonic decomposition gives the whole spectrum through the following standard identity.<sup>56</sup>

**Lemma 4.4.** Let  $0 \leq j \leq k \leq n/2$ . Then  $H_k^{\text{comp}}$  acts on  $\mathcal{H}_j^{(k)}$  as multiplication by  $j(n-j+1)$ .

*Proof.* Counting one-element replacements gives  $H_k^{\text{comp}} = k(n-k+1)I - R_{k-1}D_k$ . By (4.7) and the harmonic condition,  $R_{k-1}D_k$  acts on  $\mathcal{H}_j^{(k)}$  as multiplication by  $(k-j)(n-j-k+1)$ , and subtraction yields  $j(n-j+1)$ .  $\square$

#### 4.2 Zero extension and the collision term

For a symmetric stochastic matrix  $P$  on  $V$ , let  $K = I - P$  and let  $K^{(\ell)}$  act as  $K$  in coordinate  $\ell$  of  $V^j$ , so that  $\sum_{\ell=1}^j K^{(\ell)}$  is the Laplacian of  $j$  independent walks with off-diagonal rates  $P(x, y)$ . For  $B \in \Omega_j$ , write

$$r_P(B) = \sum_{\substack{x, y \in B \\ x \neq y}} P(x, y) \quad (4.10)$$

for the total rate of attempted jumps between occupied vertices; the sum is over ordered pairs.

A function  $h$  on  $\Omega_j$  induces a symmetric function on the tuples in  $V^j$  with distinct coordinates. Its extension by zero to all of  $V^j$  is

$$F(x_1, \dots, x_j) = \begin{cases} h(\{x_1, \dots, x_j\}), & x_1, \dots, x_j \text{ are distinct,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.11)$$

The next identity relates the energy of the zero extension to the exclusion energy. The additional term accounts for attempted jumps between occupied vertices and will be bounded using the heat kernel.

**Lemma 4.5.** Let  $1 \leq j \leq n/2$ , let  $h \in \mathcal{H}_j$ , and let  $F$  be its zero extension. Then  $F$  has mean zero in each coordinate with the others fixed, and

$$\|F\|_{2, \pi^{\otimes j}}^2 = \frac{(n)_j}{n^j} \|h\|_{2, \pi_j}^2. \quad (4.12)$$

For every symmetric stochastic matrix  $P$  on  $V$ ,

$$\left\langle F, \sum_{\ell=1}^j K^{(\ell)} F \right\rangle_{\pi^{\otimes j}} = \frac{(n)_j}{n^j} (\langle h, \Phi_j(I - P)h \rangle_{\pi_j} + \mathbb{E}_{\pi_j}[r_P(B)h(B)^2]). \quad (4.13)$$

<sup>56</sup> Y. Filmus. ‘An orthogonal basis for functions over a slice of the Boolean hypercube’. In: *Electron. J. Combin.* 23.1, 1.23 (2016). 27 pp., Lemma 18, proof.

*Proof.* Fix all but one coordinate, leaving a coordinate fibre of  $V^j$ . If a vertex is repeated among the fixed coordinates,  $F$  vanishes identically on this fibre. Otherwise, the fixed coordinates form a set  $S \in \Omega_{j-1}$ , and the harmonic condition gives  $\sum_{x \notin S} h(S \cup \{x\}) = 0$  when we sum over the free coordinate. Thus the zero extension has mean zero on every coordinate fibre. Since each  $j$ -element set has  $j!$  orderings, (4.12) follows as well.

For the energy identity, consider a tuple with distinct coordinates and underlying set  $B$ . Jumps to unoccupied vertices contribute  $(\Phi_j(I-P)h)(B)$  to  $\sum_{\ell=1}^j K^{(\ell)}F$ , whereas jumps to occupied vertices contribute  $r_P(B)h(B)$ . Tuples with repeated coordinates contribute nothing to the inner product because  $F$  vanishes there. Grouping the  $j!$  orderings of each set therefore gives (4.13), with the same factor  $(n)_j/n^j$  as in (4.12).  $\square$

#### 4.3 A spectral lower bound on each harmonic subspace

We now combine the zero-extension identity with the heat-kernel comparison. At a fixed multiple of the relaxation time, the operator-norm bound controls the collision rate on every set of at most  $n/2$  vertices, while the Poincaré inequality in each coordinate provides a lower bound for the energy of the zero extension.

**Theorem 4.6.** Let  $1 \leq j \leq k \leq n/2$  and  $f \in \mathcal{H}_j^{(k)}$ . Then

$$\langle f, H_k f \rangle_{\pi_k} \geq \frac{j\lambda}{4 \log 6} \|f\|_{2, \pi_k}^2. \quad (4.14)$$

*Proof.* By Lemma 4.3, it suffices to prove the bound on  $\mathcal{H}_j$ . Let  $u = (\log 6)/\lambda$  and  $P = P_u$ . We first bound the collision term for this choice of heat kernel. The spectral gap of  $I-P$  is  $5/6$ , and

$$P \preceq \frac{1}{6}I + \frac{5}{6n}\mathbf{1}\mathbf{1}^T. \quad (4.15)$$

For  $B \in \Omega_j$ , applying this inequality to its indicator vector gives

$$r_P(B) \leq \mathbf{1}_B^T P \mathbf{1}_B \leq \frac{j}{6} + \frac{5j^2}{6n} \leq \frac{7j}{12}, \quad (4.16)$$

where we used the non-negative diagonal entries of  $P$  and  $j \leq n/2$ .

Let  $\mathbf{h} \in \mathcal{H}_j$  and let  $\mathbf{F}$  be its zero extension. Since  $\mathbf{F}$  has mean zero on each coordinate fibre, we may apply the Poincaré inequality for  $\mathbf{I} - \mathbf{P}$  in each coordinate<sup>57</sup> and sum to obtain

$$\left\langle \mathbf{F}, \sum_{\ell=1}^j \mathbf{K}^{(\ell)} \mathbf{F} \right\rangle_{\pi^{\otimes j}} \geq \frac{5j}{6} \|\mathbf{F}\|_{2, \pi^{\otimes j}}^2.$$

Substituting (4.12) and (4.13) and cancelling  $(\mathbf{n})_j / \mathbf{n}^j$ , we can now subtract the collision bound (4.16) to obtain

$$\langle \mathbf{h}, \Phi_j(\mathbf{I} - \mathbf{P})\mathbf{h} \rangle_{\pi_j} \geq \left( \frac{5j}{6} - \frac{7j}{12} \right) \|\mathbf{h}\|_{2, \pi_j}^2 = \frac{j}{4} \|\mathbf{h}\|_{2, \pi_j}^2. \quad (4.17)$$

Finally, Theorem 3.4 gives  $\Phi_j(\mathbf{I} - \mathbf{P}) \preceq \mathbf{u} \mathbf{H}_j$ , so dividing by  $\mathbf{u}$  and applying (4.9) yields (4.14).  $\square$

## 5. Controlling the harmonic components of a density

To obtain a mixing estimate from Theorem 4.6, we first bound the harmonic components of densities with respect to the uniform stationary measure. We use inclusion probabilities for this purpose, conditioning the law at a fixed time on configurations which avoid vertices with large occupation probabilities. We then combine the resulting bounds with the decay on each harmonic subspace to obtain mixing on the independent-particle time scale.

### 5.1 An inclusion inequality

The occupation probabilities evolve according to the single-particle heat equation, while the classical negative correlation inequality bounds inclusion probabilities by products of occupation probabilities. We record these facts in the following proposition. For the inequality, see Liggett<sup>58</sup> and Borcea, Brändén and Liggett<sup>59</sup>.

<sup>57</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Lemma 13.7.

<sup>58</sup> T. M. Liggett. *Interacting Particle Systems*. Vol. 276. Grundlehren Math. Wiss. New York: Springer-Verlag, 1985, Chapter VIII, Proposition 1.7.

<sup>59</sup> J. Borcea, P. Brändén and T. M. Liggett. ‘Negative dependence and the geometry of polynomials’. In: *J. Amer. Math. Soc.* 22.2 (2009). Author version cited, pp. 521–567. arXiv: 0707.2340v2, equation (17).

**Proposition 5.1.** Let  $(B_t)_{t \geq 0}$  be the exclusion process started from a deterministic configuration  $B_0 \in \Omega_k$ , with occupation probabilities  $m_t(x) = \mathbb{P}_{B_0}(x \in B_t)$ . Then

$$m_t(x) = \sum_{a \in B_0} P_t(a, x). \quad (5.1)$$

For every  $S \subseteq V$  and every  $t \geq 0$ ,

$$\mathbb{P}_{B_0}(S \subseteq B_t) \leq \prod_{x \in S} m_t(x). \quad (5.2)$$

An empty product is interpreted as one.

### 5.2 Conditioning on moderate occupation probabilities

The inclusion inequality becomes a uniform bound once we exclude configurations meeting a vertex with large occupation probability. The next proposition bounds the probability of the excluded configurations in terms of  $kd_1(t)$ .

**Proposition 5.2.** Let  $1 \leq k \leq n/2$  and  $t \geq 0$  satisfy  $2kd_1(t) < 1$ , and let  $\mu$  be the law at time  $t$  of  $EX(k)$  started from a deterministic configuration. Then there is an event  $E \subseteq \Omega_k$ , of probability  $p = \mu(E) \geq 1 - 2kd_1(t) > 0$ , such that  $\nu = \mu(\cdot | E)$  satisfies

$$\nu\{B : S \subseteq B\} \leq p^{-1} \left( \frac{2k}{n} \right)^j \quad (S \in \Omega_j, \ 1 \leq j \leq k). \quad (5.3)$$

*Proof.* Let  $B_0$  be the initial configuration and write  $m(x) = \mu\{B : x \in B\}$ . Using  $\sum_x m(x) = k$  and (5.1), we have

$$\sum_x \left( m(x) - \frac{k}{n} \right)_+ = \frac{1}{2} \sum_x \left| m(x) - \frac{k}{n} \right| \leq \sum_{a \in B_0} \|P_t(a, \cdot) - \pi\|_{TV} \leq kd_1(t), \quad (5.4)$$

where  $a_+ = \max\{a, 0\}$ . Let

$$D = \{x \in V : m(x) > 2k/n\}, \quad E = \{B \in \Omega_k : B \cap D = \emptyset\}.$$

Since  $m(x) \leq 2(m(x) - k/n)$  for  $x \in D$ , the union bound and (5.4) give

$$1 - p \leq \sum_{x \in D} m(x) \leq 2kd_1(t). \quad (5.5)$$

It remains to bound the inclusion probabilities under  $\nu$ . If  $S$  meets  $D$ , the probability is zero; otherwise, Proposition 5.1 gives

$$\nu\{B : S \subseteq B\} \leq p^{-1} \mu\{B : S \subseteq B\} \leq p^{-1} \prod_{x \in S} m(x) \leq p^{-1} (2k/n)^j. \quad \square$$

### 5.3 An averaging operator and inclusion probabilities

We next relate inclusion probabilities to the quadratic form of an averaging operator whose action on the harmonic subspaces is explicit. Fix  $1 \leq k \leq n/2$ . For  $0 \leq j \leq k$ , let  $M_j$  be the transition kernel obtained by choosing uniformly a  $j$ -element subset of the current configuration, followed by a  $k$ -element superset chosen uniformly from those containing it. Thus

$$M_j(B, C) = \frac{\binom{|B \cap C|}{j}}{\binom{k}{j} \binom{n-j}{k-j}}. \quad (5.6)$$

In particular,  $M_0$  is the projection onto the constants and  $M_k$  is the identity. For each  $j$ , denote by  $\Pi_j$  the orthogonal projection in  $L^2(\pi_k)$  onto  $\mathcal{H}_j^{(k)}$ .

The chain with transition kernel  $M_j$  is a special case of the Bernoulli-Laplace down-up chain studied by Khare and Zhou.<sup>60</sup> Here each colour occurs once, and  $k-j$  is the number of balls moved at each of the two steps. Their spectral formula gives (5.7); we include a short proof using the inclusion operators, together with the identity for inclusion probabilities needed below.

**Lemma 5.3.** Let  $1 \leq k \leq n/2$  and  $0 \leq j \leq k$ . Then  $M_j$  is self-adjoint and positive semidefinite on  $L^2(\pi_k)$ . Its restriction to  $\mathcal{H}_j^{(k)}$  is

$$M_j|_{\mathcal{H}_j^{(k)}} = \frac{\binom{n-2j}{k-j}}{\binom{k}{j} \binom{n-j}{k-j}} I = \frac{1}{\binom{k}{j}} \frac{(n-k)_j}{(n-j)_j} I. \quad (5.7)$$

For any probability measure  $\nu$  on  $\Omega_k$ , with density  $f = \nu/\pi_k$ ,

$$\langle f, M_j f \rangle_{\pi_k} = \frac{\binom{n}{j}}{\binom{k}{j}^2} \sum_{S \in \Omega_j} \nu\{B : S \subseteq B\}^2. \quad (5.8)$$

*Proof.* Taking the adjoint with respect to counting measure, we have

$$M_j = \frac{U_{j,k} U_{j,k}^*}{\binom{k}{j} \binom{n-j}{k-j}}. \quad (5.9)$$

Indeed, the  $(B, C)$  entry of  $U_{j,k} U_{j,k}^*$  is the number of  $j$ -element subsets of  $B \cap C$ . The factorisation shows that  $M_j$  is self-adjoint and positive semidefinite in  $L^2(\pi_k)$ , and applying (4.4) to  $h \in \mathcal{H}_j$

---

<sup>60</sup> K. Khare and H. Zhou. 'Rates of convergence of some multivariate Markov chains with polynomial eigenfunctions'. In: *Ann. Appl. Probab.* 19.2 (2009), pp. 737-777, Definition 4.16 and Theorem 4.19(2).

gives the first equality in (5.7). The second equality follows on expanding the binomial coefficients.

To prove (5.8), substitute  $f(B) = \binom{n}{k} \nu(B)$  into the quadratic form of  $M_j$  and write

$$\binom{|B \cap C|}{j} = \sum_{S \in \Omega_j} 1_{\{S \subseteq B\}} 1_{\{S \subseteq C\}}.$$

Interchanging the sums gives (5.8), since  $\binom{n}{k} / [\binom{k}{j} \binom{n-j}{k-j}] = \binom{n}{j} / \binom{k}{j}^2$ .  $\square$

We now use the quadratic form of the averaging operator to bound each harmonic component of the density in terms of the inclusion probabilities.

**Proposition 5.4.** Let  $1 \leq k \leq n/2$ ,  $p \in (0, 1]$ , and let  $\nu$  be a probability measure on  $\Omega_k$  satisfying (5.3) for every  $1 \leq j \leq k$ . Then its density  $f = \nu/\pi_k$  satisfies

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq p^{-1} \frac{(4k)^j}{j!} \quad (1 \leq j \leq k). \quad (5.10)$$

*Proof.* Since  $M_j$  is self-adjoint and leaves  $\mathcal{H}_j^{(k)}$  invariant, the quadratic form splits over this subspace and its orthogonal complement. By positive semidefiniteness and (5.7), we therefore have

$$\frac{1}{\binom{k}{j}} \frac{(n-k)_j}{(n-j)_j} \|\Pi_j f\|_{2, \pi_k}^2 \leq \langle f, M_j f \rangle_{\pi_k}. \quad (5.11)$$

To bound the right-hand side, observe that every configuration contains exactly  $\binom{k}{j}$  subsets of size  $j$ , whence  $\sum_{S \in \Omega_j} \nu\{B : S \subseteq B\} = \binom{k}{j}$ . Thus, in each squared probability in (5.8), we bound one factor by (5.3) and sum the other to obtain

$$\langle f, M_j f \rangle_{\pi_k} \leq p^{-1} \frac{\binom{n}{j}}{\binom{k}{j}} \left( \frac{2k}{n} \right)^j.$$

Combining this with (5.11) gives

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq p^{-1} \binom{n}{j} \left( \frac{2k}{n} \right)^j \frac{(n-j)_j}{(n-k)_j}. \quad (5.12)$$

It remains to estimate the falling-factorial ratio. Each denominator factor is positive and, for  $0 \leq i < j$ , we have  $n-j-i \leq 2(n-k-i)$ , since  $n-2k+j-i \geq 0$ . Hence the ratio in (5.12) is at most  $2^j$ , and (5.10) follows using  $\binom{n}{j} \leq n^j/j!$ .  $\square$

This bound depends on the number of particles, whereas the squared norms of the harmonic components of the density of a point mass are the dimensions of the corresponding spaces; see (7.5).

#### 5.4 A two-time mixing estimate

We can now combine the preceding bounds with the spectral estimate. We condition the law at time  $t$  as above, then evolve the resulting conditional law for a further time  $s$ .

**Proposition 5.5.** Let  $1 \leq k \leq n/2$  and  $t, s \geq 0$ . Then

$$d_{\text{EX}(k)}(t+s) \leq 2kd_1(t) + \frac{1}{2} \left[ \exp \left( 4ke^{-\lambda s/(2 \log 6)} \right) - 1 \right]^{1/2}. \quad (5.13)$$

*Proof.* The assertion is immediate if  $2kd_1(t) \geq 1$ , so suppose otherwise. Fix an initial configuration and let  $\mu$  be the law at time  $t$ . By Proposition 5.2, we may write  $\mu = p\nu + (1-p)\beta$ , where  $p \geq 1 - 2kd_1(t) > 0$ ,  $\nu$  satisfies (5.3), and  $\beta$  is a probability measure, chosen arbitrarily if  $p = 1$ .

Let  $f = \nu/\pi_k$ . We have  $\Pi_0 f = 1$ , since  $\nu$  is a probability measure, and by reversibility its density after a further time  $s$  is  $e^{-sH_k}f$ . The harmonic subspaces are orthogonal and invariant, so Theorem 4.6 gives

$$\begin{aligned} \|e^{-sH_k}(f-1)\|_{2,\pi_k}^2 &= \sum_{j=1}^k \|e^{-sH_k}\Pi_j f\|_{2,\pi_k}^2 \\ &\leq \sum_{j=1}^k \exp\left(-\frac{j\lambda s}{2 \log 6}\right) \|\Pi_j f\|_{2,\pi_k}^2. \end{aligned} \quad (5.14)$$

Using  $\mu = p\nu + (1-p)\beta$ , the  $L^2$  bound on total variation distance<sup>61</sup>, and Proposition 5.4, we obtain

$$\begin{aligned} \|\mu Q_s^{(k)} - \pi_k\|_{\text{TV}} &\leq 1-p + \frac{p}{2} \|e^{-sH_k}(f-1)\|_{2,\pi_k} \\ &\leq 2kd_1(t) + \frac{\sqrt{p}}{2} \left[ \sum_{j=1}^k \frac{(4k)^j}{j!} \exp\left(-\frac{j\lambda s}{2 \log 6}\right) \right]^{1/2}. \end{aligned} \quad (5.15)$$

The result follows by using  $p \leq 1$ , extending the sum to all  $j \geq 1$ , and summing the exponential series. All bounds are uniform in the initial configuration.  $\square$

<sup>61</sup> D. A. Levin and Y. Peres. *Markov Chains and Mixing Times*. 2nd ed. With contributions by E. L. Wilmer. Providence, RI: American Mathematical Society, 2017, Section 4.7, equation (4.37).

## 6. Proof of the main theorem

Proposition 5.5 gives a mixing-time bound with an additive term, which we compare with  $t_1(\varepsilon/k)$  using the spectral lower bound. For at least two particles, this controls the additional time  $s$  except when  $k=2$  and  $\varepsilon$  is close to one; we complete the proof in that range by truncating the single-particle transition kernel.

### 6.1 An error-dependent additive bound

**Proposition 6.1.** Let  $1 \leq k \leq n/2$ ,  $0 < \varepsilon < 1$  and  $\tau \geq 0$  satisfy  $d_1(\tau) \leq \varepsilon/(4k)$ . Then

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq \tau + \frac{2 \log 6}{\lambda} \log \frac{8k}{\varepsilon^2}. \quad (6.1)$$

In particular,

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq t_1\left(\frac{\varepsilon}{4k}\right) + \frac{2 \log 6}{\lambda} \log \frac{8k}{\varepsilon^2}. \quad (6.2)$$

*Proof.* Apply Proposition 5.5 with  $t = \tau$  and  $s = (2 \log 6) \lambda^{-1} \log(8k/\varepsilon^2)$ . By the assumption on the single-particle distance, the first term is at most  $\varepsilon/2$ . The argument of the outer exponential in the second term is  $\varepsilon^2/2$ , so the inequality  $e^x - 1 \leq 2x$  for  $0 \leq x \leq 1/2$  bounds this term by  $\varepsilon/2$  as well. Hence  $d_{\text{EX}(k)}(\tau + s) \leq \varepsilon$ , and the last assertion follows by taking  $\tau = t_1(\varepsilon/(4k))$ .  $\square$

We next compare both terms in the additive bound with the single-particle mixing time at the error appearing in the main theorem.

**Proposition 6.2.** Let  $2 \leq k \leq n/2$ . Then for  $0 < \varepsilon \leq 1/2$ ,

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq (3 + 12 \log 6) t_1(\varepsilon/k) < 25 t_1(\varepsilon/k). \quad (6.3)$$

If  $3 \leq k \leq n/2$ , then for every  $0 < \varepsilon < 1$ ,

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq (5 + 16 \log 6) t_1(\varepsilon/k) < 34 t_1(\varepsilon/k). \quad (6.4)$$

*Proof.* Let  $T = t_1(\varepsilon/k)$ . Since  $\varepsilon/k < 1/2$  in both ranges, Lemma 2.3 gives

$$d_1(mT) \leq \frac{\varepsilon}{k} \left( \frac{2\varepsilon}{k} \right)^{m-1}, \quad \lambda T \geq \log \frac{k}{2\varepsilon}. \quad (6.5)$$

If  $k \geq 2$  and  $\varepsilon \leq 1/2$ , then  $2\varepsilon/k \leq 1/2$  and hence  $d_1(3T) \leq \varepsilon/(4k)$ . Furthermore,

$$\begin{aligned} \log \frac{8k}{\varepsilon^2} &= 2 \log \frac{k}{2\varepsilon} + \log \frac{32}{k} \\ &\leq 2 \log \frac{k}{2\varepsilon} + \log 16 \leq 6 \log \frac{k}{2\varepsilon}, \end{aligned}$$

where the last step uses  $k/(2\varepsilon) \geq 2$ . We may therefore apply Proposition 6.1 with  $\tau = 3T$ ; the second inequality in (6.5) then gives (6.3).

Now suppose that  $k \geq 3$  and  $0 < \varepsilon < 1$ . Using  $2\varepsilon/k \leq 2/3$  and  $(2/3)^4 < 1/4$ , we obtain  $d_1(5T) \leq \varepsilon/(4k)$ . To bound the additional time, we use the inequality

$$\log \frac{8k}{\varepsilon^2} \leq 8 \log \frac{k}{2\varepsilon} \quad (6.6)$$

which is equivalent to  $2048\varepsilon^6 \leq k^7$  and holds because  $2048\varepsilon^6 < 2048 < 3^7 \leq k^7$ . Applying Proposition 6.1 with  $\tau = 5T$ , followed by (6.6) and (6.5), gives (6.4). The numerical comparisons follow from  $3 + 12 \log 6 < 25$  and  $5 + 16 \log 6 < 34$ .  $\square$

## 6.2 A density bound for two particles

When  $k = 2$ , the right-hand side of the lower bound  $\lambda t_1(\varepsilon/2) \geq \log(1/\varepsilon)$  tends to zero as  $\varepsilon \uparrow 1$ . We therefore use a different estimate in this range. The following estimate for the single-particle transition kernel will allow us to find a subprobability measure whose density with respect to the uniform stationary measure is bounded and whose total mass is bounded away from zero.

For a matrix  $R$  on  $V$ , write  $Q(x, y) = (R(x, y) - 1/n)_+$ .

**Lemma 6.3.** Let  $R$  be a doubly stochastic matrix on  $V$  such that

$$\max_{x \in V} \frac{1}{2} \sum_{y \in V} |R(x, y) - 1/n| \leq \frac{1}{2}.$$

Then for every integer  $m \geq 1$ ,

$$R^m(x, y) \leq \frac{2}{n} + Q^m(x, y), \quad \sum_y Q^m(x, y) \leq 2^{-m}. \quad (6.7)$$

*Proof.* Since  $Q$  is non-negative with row sums at most  $1/2$ , the row sums of  $Q^m$  are at most  $2^{-m}$ . To prove the remaining inequality, let  $B = R - Q$ ; then  $0 \leq B(x, y) \leq 1/n$  entrywise, and the telescoping identity

$$R^m - Q^m = \sum_{i=0}^{m-1} Q^i B R^{m-1-i} \quad (6.8)$$

holds for arbitrary matrices  $R$  and  $Q$ . As  $R$  is doubly stochastic, we have  $(BR^\ell)(x, y) \leq 1/n$  for every  $\ell \geq 0$ . Together with the bound on the row sums of  $Q^i$ , this shows that the  $i$ th summand in (6.8) is at most  $2^{-i}/n$  entrywise. Summing the geometric series proves the first inequality in (6.7).  $\square$

The preceding estimate gives the required subprobability measure, from which we obtain bounds on the total variation distance from stationarity before and after further evolution.

**Lemma 6.4.** Let  $n \geq 4$  and  $T \geq 0$  satisfy  $d_1(T) \leq 1/2$ , and let  $\mu$  be the law at time  $6T$  of the two-particle exclusion process started from a deterministic configuration. Then there is a non-negative measure  $g \leq \mu$  on  $\Omega_2$  such that

$$g(\Omega_2) \geq \frac{27}{32}, \quad g \leq \frac{25}{2}\pi_2. \quad (6.9)$$

Consequently, uniformly in the initial configuration,

$$d_{\text{EX}(2)}(6T) \leq \frac{373}{400}, \quad (6.10)$$

$$d_{\text{EX}(2)}(6T + s) \leq \frac{5}{32} + \frac{5}{2\sqrt{2}}e^{-\lambda s} \quad (s \geq 0). \quad (6.11)$$

*Proof.* Let the initial particles be at  $\mathbf{a}$  and  $\mathbf{b}$ . Applying Lemma 6.3 to  $R = P_T$  with  $m = 6$ , we find that the occupation probabilities at time  $6T$  satisfy

$$m(\mathbf{y}) = P_{6T}(\mathbf{a}, \mathbf{y}) + P_{6T}(\mathbf{b}, \mathbf{y}) \leq \frac{4}{n} + u(\mathbf{y}), \quad \sum_{\mathbf{y}} u(\mathbf{y}) \leq \frac{1}{32},$$

where  $u(\mathbf{y}) = Q^6(\mathbf{a}, \mathbf{y}) + Q^6(\mathbf{b}, \mathbf{y}) \geq 0$ . Let  $D = \{\mathbf{y} : m(\mathbf{y}) > 5/n\}$  and  $E = \{B \in \Omega_2 : B \cap D = \emptyset\}$ . Since  $\mathbf{y} \in D$  implies  $u(\mathbf{y}) \geq m(\mathbf{y}) - 4/n > m(\mathbf{y})/5$ , the union bound gives

$$\mu(E^c) \leq \sum_{\mathbf{y} \in D} m(\mathbf{y}) \leq 5 \sum_{\mathbf{y}} u(\mathbf{y}) \leq \frac{5}{32}. \quad (6.12)$$

Let  $g(C) = \mu(C)1_E(C)$ . For  $C = \{\mathbf{x}, \mathbf{y}\} \in E$ , Proposition 5.1 gives  $g(C) \leq m(\mathbf{x})m(\mathbf{y}) \leq 25/n^2$ , and since  $\pi_2(C) = 2/[n(n-1)]$ , we obtain  $g(C) \leq (25/2)\pi_2(C)$ . Together with (6.12), this proves (6.9).

Write  $p = g(\Omega_2)$ . Since  $2g/25$  is bounded by both  $\mu$  and  $\pi_2$ , their overlap is at least  $2p/25 \geq 27/400$ ; thus (6.10) follows from the identity  $\|\mu - \pi_2\|_{\text{TV}} = 1 - \sum_C \min\{\mu(C), \pi_2(C)\}$ .

To prove (6.11), let  $h = g/\pi_2$ . Since  $0 \leq h \leq 25/2$  and  $\mathbb{E}_{\pi_2} h = p$ , we have  $\|h - p\|_{2, \pi_2}^2 \leq (25/2)p - p^2 \leq 25/2$ . Since the remaining measure  $\mu - g$  has mass  $1 - p$ , its total variation distance from  $(1 - p)\pi_2$  remains at most  $1 - p$  after evolution. Theorem 2.1 and reversibility now give

$$\begin{aligned} \|\mu Q_s^{(2)} - \pi_2\|_{\text{TV}} &\leq 1 - p + \frac{1}{2} \|e^{-sH_2}(h - p)\|_{2, \pi_2} \\ &\leq \frac{5}{32} + \frac{1}{2} e^{-\lambda s} \sqrt{\frac{25}{2}}, \end{aligned}$$

which is (6.11). □

We use the two bounds in the preceding lemma to complete the comparison for two particles.

**Proposition 6.5.** Let  $n \geq 4$  and  $1/2 < \varepsilon < 1$ . Then

$$t_{\text{mix}}^{\text{EX}(2)}(\varepsilon) < 34t_1(\varepsilon/2). \quad (6.13)$$

*Proof.* Let  $T = t_1(\varepsilon/2)$ . We have  $d_1(T) \leq \varepsilon/2 < 1/2$ , so Lemma 6.4 applies. For  $\varepsilon \geq 15/16$ , we have  $373/400 < 15/16 \leq \varepsilon$  and hence  $t_{\text{mix}}^{\text{EX}(2)}(\varepsilon) \leq 6T$ . It remains to consider  $1/2 < \varepsilon < 15/16$ . Taking  $s = (\log 6)/\lambda$  in (6.11) gives

$$d_{\text{EX}(2)}(6T + s) \leq \frac{5}{32} + \frac{5}{12\sqrt{2}} < \frac{1}{2} < \varepsilon.$$

Lemma 2.3 gives  $\lambda T \geq \log(1/\varepsilon) > \log(16/15)$ , and hence

$$t_{\text{mix}}^{\text{EX}(2)}(\varepsilon) \leq 6T + \frac{\log 6}{\lambda} < \left(6 + \frac{\log 6}{\log(16/15)}\right) T < 34T.$$

The last numerical inequality is equivalent to  $6 < (16/15)^{28}$ , completing the proof.  $\square$

### 6.3 Completion of the proof

*Proof of Theorem 1.1.* By (2.6), it suffices to prove the result for  $k \leq n/2$ , in which case  $r = k$ . If  $k = 1$ , exclusion coincides with the single-particle walk, so the conclusion holds also when their mixing times vanish. For the other particle numbers, Proposition 6.2 gives (1.4) when  $k \geq 2$  and  $0 < \varepsilon \leq 1/2$ , while its second part gives (1.3) when  $k \geq 3$  and  $0 < \varepsilon < 1$ . The remaining case,  $k = 2$  and  $1/2 < \varepsilon < 1$ , follows from Proposition 6.5.  $\square$

Finally,  $\min\{k, n - k\} \leq k$ , and  $t_1$  is non-increasing in its error argument. Hence Theorem 1.1 gives  $t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq 34t_1(\varepsilon/k)$  for every  $1 \leq k < n$ , proving the original formulation (1.2) for all particle numbers.

## 7. Consequences of the comparison

We first compare the total-variation,  $L^2$  and uniform mixing times of exclusion with those of independent particles. We then prove the Dirichlet-form comparison stated in the introduction and use the spectral estimates to obtain a bound in terms of the relaxation time, a logarithmic Sobolev inequality and energy bounds for the Heisenberg ferromagnet.

### 7.1 Distances from stationarity and independent particles

**Corollary 7.1.** Let  $1 \leq k < n$  and  $r = \min\{k, n - k\}$ . Then

$$d_{\text{EX}(k)}(34t) \leq \min\{1, rd_1(t)\} \quad (t \geq 0). \quad (7.1)$$

*Proof.* The assertion is immediate if  $rd_1(t) \geq 1$ . Otherwise, let  $rd_1(t) < \varepsilon < 1$ . Since  $t_1(\varepsilon/r) \leq t$ , Theorem 1.1 gives  $t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq 34t$ , and hence  $d_{\text{EX}(k)}(34t) \leq \varepsilon$ . Letting  $\varepsilon$  decrease to  $rd_1(t)$  gives the required bound, also when the limit is zero.  $\square$

Conversely, taking  $t = t_1(\varepsilon/r)$  in (7.1) gives (1.3). Thus the comparison of mixing times is equivalent to the comparison of distances from stationarity at all times.

**Corollary 7.2.** Let  $1 \leq k < n$  and  $r = \min\{k, n - k\}$ . Then

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq 100 t_{\text{mix}}^{\text{RW}(r)}(\varepsilon) \quad (0 < \varepsilon \leq 1/4). \quad (7.2)$$

At the fixed threshold  $1/4$ , we have the sharper bound

$$t_{\text{mix}}^{\text{EX}(k)}(1/4) \leq (8 + 24 \log 6) t_{\text{mix}}^{\text{RW}(r)}(1/4) < 52 t_{\text{mix}}^{\text{RW}(r)}(1/4). \quad (7.3)$$

*Proof.* The first assertion follows by combining (1.4) with Proposition 2.4. To prove the second, we may assume  $k \leq n/2$  by particle-hole symmetry. For  $k=1$  the two processes coincide, so suppose that  $k \geq 2$  and let  $T = t_{\text{mix}}^{\text{RW}(k)}(1/4)$ . By Proposition 2.5, we have  $d_1(8T) \leq 1/(64k)$ ; applying Proposition 5.5 with  $s = (2 \log 6) \lambda^{-1} \log(32k)$  therefore gives

$$d_{\text{EX}(k)}(8T + s) \leq \frac{1}{32} + \frac{1}{2} \sqrt{e^{1/8} - 1} < \frac{1}{4}.$$

Hence the mixing time is at most  $8T + (2 \log 6) \lambda^{-1} \log(32k)$ . Dividing by  $T$  and using the bound  $T \geq \log(2k)/(4\lambda)$  from Proposition 2.5, together with  $\log(32k) \leq 3 \log(2k)$  for  $k \geq 2$ , we obtain (7.3).  $\square$

### 7.2 $L^2$ and uniform mixing times

The bounds on inclusion probabilities and the harmonic spectral estimate also give a comparison in  $L^2$ . The corresponding comparison for uniform mixing times follows by reversibility.

For any of the processes  $X$  considered here, let  $Q_t^X$  and  $\pi^X$  denote its transition kernel and stationary measure. We write

$$d_p^X(t) = \max_z \left\| \frac{Q_t^X(z, \cdot)}{\pi^X} - 1 \right\|_{L^p(\pi^X)} \quad (p \in \{2, \infty\}),$$

and let  $t_{\text{mix}}^{X,p}(\varepsilon)$  be the least  $t \geq 0$  for which  $d_p^X(t) \leq \varepsilon$ . Thus  $d_2^X(t)^2$  is the maximum  $\chi^2$  divergence from stationarity, and  $d_\infty^X(t)$  is the maximum relative error.

**Corollary 7.3.** Let  $c_{xy} = c_{yx} \geq 0$  be edge rates with connected positive support on a set  $V$  of  $n \geq 2$  vertices. Let  $1 \leq k < n$  and  $r = \min\{k, n - k\}$ . Then, for  $p \in \{2, \infty\}$ ,

$$t_{\text{mix}}^{\text{EX}(k),p}(\varepsilon) \leq 24 t_{\text{mix}}^{\text{RW}(r),p}(\varepsilon) \quad (0 < \varepsilon \leq 1).$$

Both processes use the edge rates  $c_{xy}$ .

*Proof.* By particle-hole symmetry, it suffices to consider  $k \leq n/2$ , and the case  $k=1$  is immediate. Suppose that  $k \geq 2$ , and let  $T = t_{\text{mix}}^{\text{RW}(k),2}(\varepsilon)$  and  $\delta = (1 + \varepsilon^2)^{1/k} - 1$ . The product formula for the  $L^2$  distance gives  $d_2^{\text{RW}(1)}(T)^2 \leq \delta$ . The spectral expansion also gives  $e^{-2\lambda T} \leq \delta$ , and  $0 < \delta \leq \sqrt{2} - 1$ .

Start exclusion from a fixed configuration, and let  $f$  be the density of its law at time  $2T$  with respect to  $\pi_k$ . Reversibility and the Cauchy-Schwarz inequality give  $nP_{2T}(x, y) \leq 1 + \delta < 2$ . Proposition 5.1 therefore bounds the inclusion probability of each  $j$ -element set by  $(2k/n)^j$ . Applying Proposition 5.4 with its parameter  $p$  equal to one, we obtain

$$\|\Pi_j f\|_{2, \pi_k}^2 \leq \frac{(4k)^j}{j!} \quad (1 \leq j \leq k).$$

Let  $s = 12(\log 6)T$ . By (5.14),

$$\begin{aligned} \|e^{-sH_k} f - 1\|_{2, \pi_k}^2 &\leq \sum_{j=1}^k \frac{(4k)^j}{j!} e^{-j\lambda s/(2 \log 6)} \\ &\leq e^{4k\delta^3} - 1 \leq (1 + \delta)^k - 1 = \varepsilon^2. \end{aligned}$$

For the last inequality, we used  $4u^3 \leq \log(1+u)$  for  $0 \leq u \leq \sqrt{2} - 1$ : indeed,  $\log(1+u) \geq u/(1+u)$  and  $4u^2(1+u) \leq 12\sqrt{2} - 16 < 1$ . The bounds are uniform in the initial configuration, and  $2T + s < 24T$ , proving the assertion for  $p=2$ .

For a finite reversible chain, the semigroup property and the Cauchy-Schwarz inequality give  $d_\infty^X(2t) = d_2^X(t)^2$ , with the maximum defining  $d_\infty^X(2t)$  attained on the diagonal. Hence  $t_{\text{mix}}^{X,\infty}(\varepsilon) = 2t_{\text{mix}}^{X,2}(\sqrt{\varepsilon})$ , and the case  $p = \infty$  follows with the same constant.  $\square$

### 7.3 Comparison with the complete graph

We obtain the comparison of Dirichlet forms by applying the spectral estimate on each harmonic subspace. Recall that  $H_k^{\text{comp}}$  is the Laplacian of exclusion on the complete graph with unit edge rates.

*Proof of Theorem 1.2.* Suppose first that  $k \leq n/2$ . Both operators preserve the orthogonal decomposition in (4.6). On the subspace of degree  $j$ , Theorem 4.6 bounds  $H_k$  below by  $j\lambda/(4\log 6)$ , whereas Lemma 4.4 shows that  $H_k^{\text{comp}}$  acts as  $j(n-j+1)I$ . The latter eigenvalue is at most  $nj$  for  $j \geq 1$ , and both operators vanish on constants; summing over the orthogonal subspaces therefore gives (1.6). The case  $k > n/2$  follows by applying the complement map, which is an isometry intertwining the  $k$ -particle and  $(n-k)$ -particle Laplacians for both graphs.  $\square$

#### 7.4 A mixing-time bound in terms of the relaxation time

For the density of a point mass, symmetry determines the norm of each harmonic component exactly. Applying the spectral estimate to these components gives the following bound in terms of the relaxation time.

**Corollary 7.4.** Let  $1 \leq k < n$  and  $0 < \varepsilon < 1$ . Then

$$t_{\text{mix}}^{\text{EX}(k)}(\varepsilon) \leq \frac{2\log 6}{\lambda} \log \frac{n}{\varepsilon^2}. \quad (7.4)$$

At error  $1/4$ , the right-hand side may be replaced by  $(2\log 6)\lambda^{-1} \log(8n)$ .

*Proof.* By particle-hole symmetry, we may assume that  $k \leq n/2$ . For  $B \in \Omega_k$ , let  $f_B$  be the density of the point mass at  $B$  with respect to  $\pi_k$ . Since the harmonic projection  $\Pi_j$  commutes with every permutation of  $V$ , its diagonal entries are equal. Their sum is the rank of  $\Pi_j$ , giving the projection identity used by Hermon, Kozma and Salez.<sup>62</sup> Lemma 4.2 therefore gives

$$\|\Pi_j f_B\|_{2, \pi_k}^2 = \dim \mathcal{H}_j^{(k)} = \binom{n}{j} - \binom{n}{j-1}. \quad (7.5)$$

Applying (5.14) to  $f_B$ , with the dimension bounded by  $n^j/j!$ , gives

$$d_{\text{EX}(k)}(t) \leq \frac{1}{2} \left[ \exp \left( n e^{-\lambda t / (2\log 6)} \right) - 1 \right]^{1/2}. \quad (7.6)$$

For  $t = (2\log 6)\lambda^{-1} \log(n/\varepsilon^2)$ , the argument of the outer exponential is  $\varepsilon^2$ . The convexity bound  $e^x - 1 \leq (e-1)x < 2x$  for  $0 < x \leq 1$  now shows that the right-hand side of (7.6) is less than  $\varepsilon$ , as required. Taking  $t = (2\log 6)\lambda^{-1} \log(8n)$  instead makes the same argument  $1/8$ , giving a distance strictly less than  $1/4$ .  $\square$

<sup>62</sup> J. Hermon, G. Kozma and J. Salez. *Universality of cutoff for the exclusion process*. Preprint, version 1. 2026. arXiv: 2609.31187v1, proof of Corollary 5, equations (70)–(72).

Comparing (7.5) with (5.10) explains the role of conditioning in the total-variation proof: it replaces the dependence on  $\mathbf{n}$  in the bounds for the harmonic components by a dependence on  $\mathbf{k}$ .

### 7.5 A logarithmic Sobolev inequality

We follow the comparison argument in Hermon and Salez<sup>63</sup>, using our complete-graph comparison to transfer the logarithmic Sobolev inequality for the Bernoulli-Laplace process to arbitrary symmetric rates. For a non-negative function  $g$  on  $\Omega_{\mathbf{k}}$ , let  $\text{Ent}_{\pi_{\mathbf{k}}}(g) = \mathbb{E}_{\pi_{\mathbf{k}}}[g \log g] - \mathbb{E}_{\pi_{\mathbf{k}}}[g] \log \mathbb{E}_{\pi_{\mathbf{k}}}[g]$ , with  $0 \log 0 = 0$ . We use the logarithmic Sobolev constant

$$\alpha_{\mathbf{k}} = \inf_{\text{Ent}_{\pi_{\mathbf{k}}}(f^2) > 0} \frac{\langle f, H_{\mathbf{k}} f \rangle_{\pi_{\mathbf{k}}}}{\text{Ent}_{\pi_{\mathbf{k}}}(f^2)}.$$

For a fixed Dirichlet form, this constant is the reciprocal of the one used by Salez<sup>64</sup>. Let  $\alpha_{\mathbf{k}}^{\text{comp}}$  be the constant corresponding to  $H_{\mathbf{k}}^{\text{comp}}$ . Salez assigns rate  $1/\mathbf{n}$  to each edge of the complete graph, so our choice of unit edge rates gives  $\alpha_{\mathbf{k}}^{\text{comp}} = \mathbf{n}/\tau_{1s}(\mathbf{k}, \mathbf{n} - \mathbf{k})$ . Lee and Yau also use edge rates  $1/\mathbf{n}$ . Rescaling their inequality<sup>65</sup>, in the form stated by Salez<sup>66</sup>, we therefore obtain

$$\alpha_{\mathbf{k}}^{\text{comp}} \geq \frac{\mathbf{n} \log 2}{2 \log(\mathbf{n}^2 / [\mathbf{k}(\mathbf{n} - \mathbf{k})])}. \quad (7.7)$$

Theorem 1.2 gives  $\alpha_{\mathbf{k}} \geq \lambda \alpha_{\mathbf{k}}^{\text{comp}} / (4\mathbf{n} \log 6)$ . For the upper bound, let  $g$  be a mean-zero eigenfunction of  $H_{\mathbf{k}}$  with eigenvalue  $\lambda$ .

Substituting  $f = 1 + \delta g$  into the defining quotient and letting  $\delta \rightarrow 0$ , we use the expansion  $\text{Ent}_{\pi_{\mathbf{k}}}((1 + \delta g)^2) = 2\delta^2 \|g\|_{2, \pi_{\mathbf{k}}}^2 + o(\delta^2)$  to obtain  $\alpha_{\mathbf{k}} \leq \lambda/2$ .

<sup>63</sup> J. Hermon and J. Salez. ‘The interchange process on high-dimensional products’. In: *Ann. Appl. Probab.* 31.1 (2021). Author version cited, pp. 84–98. arXiv: 1905.02146v2, Section 2.3, equation (30).

<sup>64</sup> J. Salez. ‘A sharp log-Sobolev inequality for the multislice’. In: *Ann. H. Lebesgue* 4 (2021), pp. 1143–1161, Section 1.2.

<sup>65</sup> T.-Y. Lee and H.-T. Yau. ‘Logarithmic Sobolev inequality for some models of random walks’. In: *Ann. Probab.* 26.4 (1998), pp. 1855–1873, Theorem 5.

<sup>66</sup> J. Salez. ‘A sharp log-Sobolev inequality for the multislice’. In: *Ann. H. Lebesgue* 4 (2021), pp. 1143–1161, Theorem 1.6.

Combining Theorems 1.2 and 2.1 with (7.7), we obtain the following result.

**Corollary 7.5.** Let  $c_{xy} = c_{yx} \geq 0$  have connected positive support on a set of  $n \geq 2$  vertices, and let  $1 \leq k < n$ . Then

$$\frac{\lambda \log 2}{8 \log 6 \log(n^2/[k(n-k)])} \leq \alpha_k \leq \frac{\lambda}{2}. \quad (7.8)$$

In particular, for each  $0 < \rho \leq 1/2$ , the ratio  $\alpha_k/\lambda$  is bounded above and below by positive constants depending only on  $\rho$  whenever  $\rho n \leq k \leq (1 - \rho)n$ .

When  $k = n/2$ , the lower bound becomes  $\lambda/(16 \log 6)$ . Thus Corollary 7.5 establishes the balanced two-colour conclusion discussed by Salez after Corollary 2.6<sup>67</sup> for every connected weighted graph.

### 7.6 The Heisenberg ferromagnet and quantum-cut energies

We apply the spectral comparisons to the spin-1/2 Heisenberg ferromagnet, using its classical correspondence with the exclusion process<sup>68</sup>.

Let  $T_{xy}$  be the operator on  $(\mathbb{C}^2)^{\otimes V}$  that interchanges the tensor factors at  $x$  and  $y$ , and put  $\mathcal{L}_G = \sum_{x < y} c_{xy}(I - T_{xy})$ . On the invariant subspace with exactly  $k$  spins in the second basis state, the restriction of  $\mathcal{L}_G$  is  $H_k$ . Hence  $\mathcal{L}_G = \bigoplus_{k=0}^n H_k$ , with trivial restrictions at  $k=0, n$ , and applying Theorem 1.2 to each summand gives

$$\mathcal{L}_G \succeq \frac{\lambda}{4n \log 6} \mathcal{L}_{K_n}. \quad (7.9)$$

The restriction to the single-particle subspace shows that the coefficient cannot exceed  $\lambda/n$ , so this comparison is optimal up to a universal constant.

In the normalisation used for quantum cuts by Basu, Brakensiek, Kothari and Putterman<sup>69</sup>, the two-site term  $I - T_{xy}$  is twice the orthogonal projection onto the two-spin singlet subspace.

<sup>67</sup> J. Salez. ‘A sharp log-Sobolev inequality for the multislice’. In: *Ann. H. Lebesgue* 4 (2021), pp. 1143–1161.

<sup>68</sup> S. Alexander and T. Holstein. ‘Lattice diffusion and the Heisenberg ferromagnet’. In: *Phys. Rev. B* 18.1 (1978), pp. 301–302; B. Nachtergaele and S. Starr. ‘Ordering of energy levels in Heisenberg models and applications’. In: *Mathematical Physics of Quantum Mechanics*. Ed. by J. Asch and A. Joye. Vol. 690. Lecture Notes in Phys. Author version cited. Berlin: Springer, 2006, pp. 149–170. arXiv: math-ph/0503056v1.

<sup>69</sup> A. Basu et al. *Quantum cut sparsifiers*. Preprint. 2026. arXiv: 2606.09728v1, Section 1.2.

Their proof of quantum-cut sparsification uses a complete-graph comparison for these operators.<sup>70</sup> Inequality (7.9) gives such a comparison with a coefficient of order  $\lambda/n$ , uniformly over all quantum states, and hence with the optimal dependence on the single-particle spectral gap.

To express the result in the conventional Heisenberg normalisation, let  $S_x^a$  be one half of the  $a$ th Pauli matrix acting at  $x$ , for  $a \in \{1, 2, 3\}$ , and write  $S_x \cdot S_y = \sum_{a=1}^3 S_x^a S_y^a$ . The ferromagnetic Hamiltonian<sup>71</sup> with ground-state energy shifted to zero is

$$\mathcal{F}_G = \sum_{x < y} c_{xy} \left( \frac{1}{4} I - S_x \cdot S_y \right) = \frac{1}{2} \mathcal{L}_G. \quad (7.10)$$

The squared total-spin operator is  $S_{\text{tot}}^2 = \sum_{a=1}^3 (\sum_x S_x^a)^2$ . The Hamiltonian preserves the eigenspaces of this operator.<sup>72</sup> For  $s \in \{n/2, n/2 - 1, \dots, n/2 - \lfloor n/2 \rfloor\}$ , write  $E_G(s)$  for the least eigenvalue of  $\mathcal{F}_G$  on the eigenspace of  $S_{\text{tot}}^2$  with eigenvalue  $s(s+1)$ . The identity

$$\mathcal{L}_{K_n} = \frac{n(n+2)}{4} I - S_{\text{tot}}^2$$

together with Lemma 4.4 identifies total spin  $n/2 - j$  with harmonic degree  $j$  in each subspace with a fixed number of particles. Applying Theorem 4.6 and particle-hole symmetry, and using the normalisation in (7.10), we obtain

$$E_G(n/2 - j) \geq \frac{j\lambda}{8 \log 6} \quad (0 \leq j \leq \lfloor n/2 \rfloor). \quad (7.11)$$

<sup>70</sup> A. Basu et al. *Quantum cut sparsifiers*. Preprint. 2026. arXiv: 2606.09728v1, Theorem 5.5.

<sup>71</sup> B. Nachtergaele and S. Starr. ‘Ordering of energy levels in Heisenberg models and applications’. In: *Mathematical Physics of Quantum Mechanics*. Ed. by J. Asch and A. Joye. Vol. 690. Lecture Notes in Phys. Author version cited. Berlin: Springer, 2006, pp. 149–170. arXiv: math-ph/0503056v1, Section 5.3, Proposition 2.

<sup>72</sup> B. Nachtergaele and S. Starr. ‘Ordering of energy levels in Heisenberg models and applications’. In: *Mathematical Physics of Quantum Mechanics*. Ed. by J. Asch and A. Joye. Vol. 690. Lecture Notes in Phys. Author version cited. Berlin: Springer, 2006, pp. 149–170. arXiv: math-ph/0503056v1, Section 1.

## 8. Failure of the same-error comparison at large errors

The comparison with independent particles at the same total-variation error, stated by Hermon and Pymar<sup>73</sup>, fails at large errors when arbitrary symmetric edge rates are allowed. We exhibit a family of connected weighted four-cycles with equal total jump rates at every vertex.

**Proposition 8.1.** Let  $\varepsilon \in [3/4, 5/6]$ . Then there is no finite constant  $C_\varepsilon$  such that

$$t_{\text{mix}}^{\text{EX}(2)}(\varepsilon) \leq C_\varepsilon t_{\text{mix}}^{\text{RW}(2)}(\varepsilon) \quad (8.1)$$

on every finite connected symmetric weighted graph, even when restricted to weighted four-cycles with equal total jump rates at all vertices.

*Proof.* For  $\eta > 0$ , let  $V = \{0, 1\}^2$  and assign the edge rates

$$c_{(a,b),(a,1-b)} = 1, \quad c_{(a,b),(1-a,b)} = \eta \quad (a, b \in \{0, 1\}). \quad (8.2)$$

The graph is connected, with total jump rate  $1 + \eta$  at every vertex; we use a subscript  $\eta$  to indicate dependence on the rates.

Start exclusion from  $B_0 = \{(0, 0), (0, 1)\}$ . Since both ends of the rate-one edge are occupied, a transition from  $B_0$  must use one of the two rate- $\eta$  edges. No such transition occurs before time  $t$  with probability  $e^{-2\eta t}$ , so  $\mathbb{P}_{B_0}(B_t = B_0) \geq e^{-2\eta t}$ . The stationary measure is uniform on six configurations; testing the singleton event  $\{B_0\}$ , of stationary probability  $1/6$ , gives

$$d_{\text{EX}(2), \eta}(t) \geq e^{-2\eta t} - \frac{1}{6}, \quad t_{\text{mix}}^{\text{EX}(2), \eta}(\varepsilon) \geq \frac{1}{2\eta} \log \frac{1}{\varepsilon + 1/6}. \quad (8.3)$$

Since  $\varepsilon < 5/6$ , the lower bound for the mixing time is positive.

For the independent process, let  $t_0 = (\log 2)/2$ . First set  $\eta = 0$ , measuring distance from the uniform distribution on  $V^2$ . For each particle, the first coordinate remains fixed, whereas the second flips at rate one, taking its initial value at time  $t_0$  with probability  $3/4$  and its other value with probability  $1/4$ . From any initial ordered pair, including coinciding positions, the law is therefore supported on four states, each of probability at least  $1/16$ . Its overlap with the uniform distribution on the sixteen states is  $4/16$ , giving total variation distance exactly  $3/4$ .

For  $\eta > 0$ , first- and second-coordinate flips commute, so the independent transition kernel is the product of the auxiliary  $\eta = 0$

<sup>73</sup> J. Hermon and R. Pymar. ‘The exclusion process mixes (almost) faster than independent particles’. In: *Ann. Probab.* 48.6 (2020), pp. 3077–3123, equation (2).

kernel and a first-coordinate flip kernel. The latter preserves the uniform measure and contracts total variation, giving

$$d_{\text{RW}(2),\eta}(t_0) \leq \frac{3}{4} \leq \varepsilon, \quad t_{\text{mix}}^{\text{RW}(2),\eta}(\varepsilon) \leq \frac{\log 2}{2}. \quad (8.4)$$

The independent mixing time is positive throughout the stated range of errors. Combining (8.3) and (8.4) yields

$$\frac{t_{\text{mix}}^{\text{EX}(2),\eta}(\varepsilon)}{t_{\text{mix}}^{\text{RW}(2),\eta}(\varepsilon)} \geq \frac{\log(1/(\varepsilon + 1/6))}{\eta \log 2} \rightarrow \infty \quad (\eta \downarrow 0). \quad \square$$

In particular, at  $\varepsilon = 4/5$ , the exclusion mixing time is at least  $(2\eta)^{-1} \log(30/29)$ , whereas the independent mixing time is at most  $(\log 2)/2$ . The difference comes from the stationary probabilities of the sets accessible along the fast edges. With the first coordinates fixed, the independent process can reach four states whose total stationary probability is  $1/4$ , while exclusion remains in its initial configuration, of stationary probability  $1/6$ .

The error scaling in Theorem 1.1 accounts for this slow coordinate. With first-coordinate flips suppressed, a single particle stays at distance at least  $1/2$  from stationarity; since  $\varepsilon/2 < 1/2$ , the time  $t_1(\varepsilon/2)$  still depends on the slow transitions.